

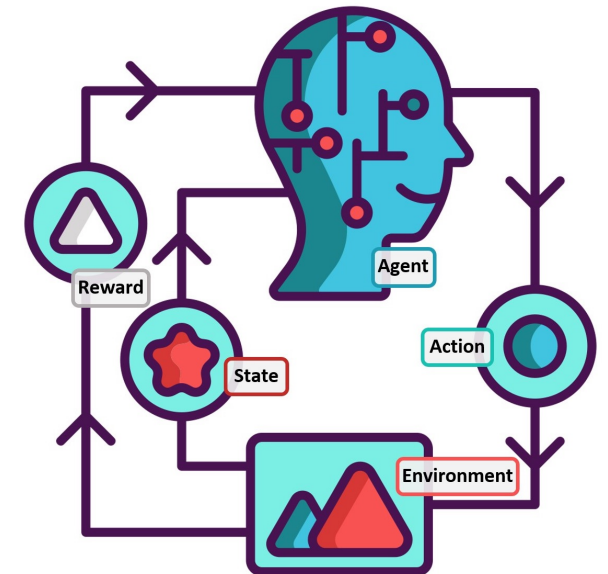
Lecture #04

Markov Decision

Processes & Bellman

Equations

Gian Antonio Susto



Announcements before starting

- The TA forgot to record the labs last week... :facepalm:
- We have uploaded last year recordings
- Please remind the TA to record the lab!
- Next lab tomorrow at 14:30 on bandits

Recap: Markov Decision Processes (MDPs)

Markov Decision Processes (MDPs)

formally describe an environment for Reinforcement Learning

- i. Markov Processes $\langle \mathcal{S}, \mathcal{P} \rangle$
- ii. Markov Reward Processes $\langle \mathcal{S}, \mathcal{P}, \mathcal{R}, \gamma \rangle$
- iii. Markov Decision Processes $\langle \mathcal{S}, \mathcal{A}, \mathcal{P}, \mathcal{R}, \gamma \rangle$

- We have seen the formal introduction of state $\mathcal{S}$, transition probability $\mathcal{P}$
- We will see other elements (the reward function $\mathcal{R}$, return G , the value function v , the value function q , the discount factor γ , the action space $\mathcal{A}$)

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MDPs are underlying formalization of a RL problem, but some of the elements ($\mathcal{P}, \mathcal{R}$) will not be known by the agent

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$$\mathcal{P}_{ss'} = \mathbb{P} [S_{t+1} = s' \mid S_t = s]$$

$$\mathcal{P} = \begin{matrix} & \text{to} \\ \text{from} & \begin{bmatrix} \mathcal{P}_{11} & \dots & \mathcal{P}_{1n} \\ \vdots & & \\ \mathcal{P}_{n1} & \dots & \mathcal{P}_{nn} \end{bmatrix} \end{matrix}$$

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$$\mathcal{R}_s = \mathbb{E} [R_{t+1} \mid S_t = s]$$

ii. Markov Reward Processes

Let's add rewards: a Markov Reward process is a Markov Chain with reward values

Definition

A **Markov Reward Process** is a tuple $\langle \mathcal{S}, \mathcal{P}, \mathcal{R}, \gamma \rangle$ such that:

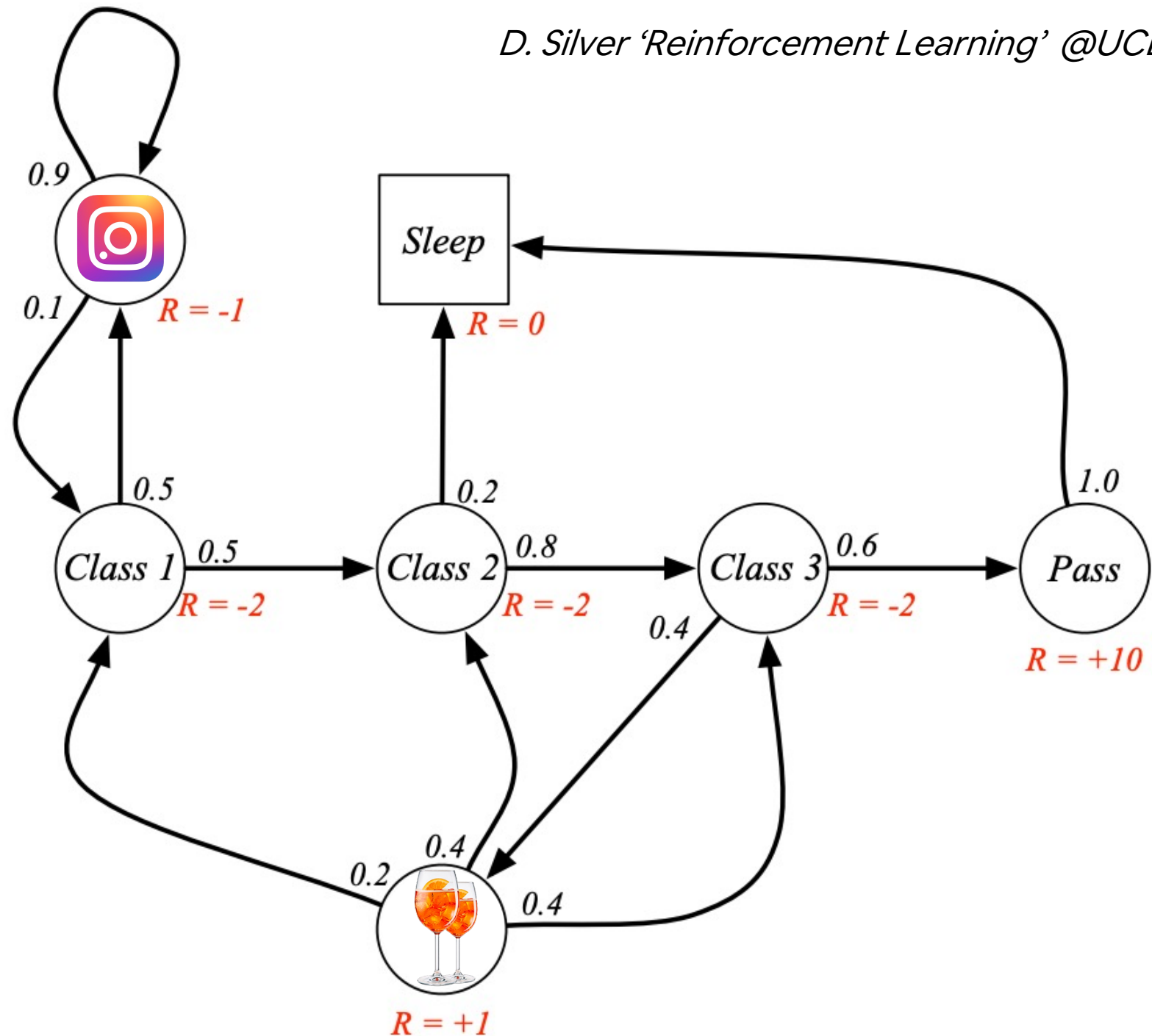
- $\mathcal{S}$ is a finite set of states
- $\mathcal{P}$ is a state transition probability matrix with entries

$$\mathcal{P}_{ss'} = \mathbb{P} [S_{t+1} = s' \mid S_t = s]$$

- $\mathcal{R}$ is a reward function, $\mathcal{R}_s = \mathbb{E} [R_{t+1} | S_t = s]$ (it is just the immediate reward, in that specific state)
- γ is a discount factor, $\gamma \in [0,1]$

ii. Markov Reward Processes : Student Markov Chain

D. Silver 'Reinforcement Learning' @UCL



ii. Markov Reward Processes: Return

Definition

The **Return** G_t is the total discounted reward from time-step t

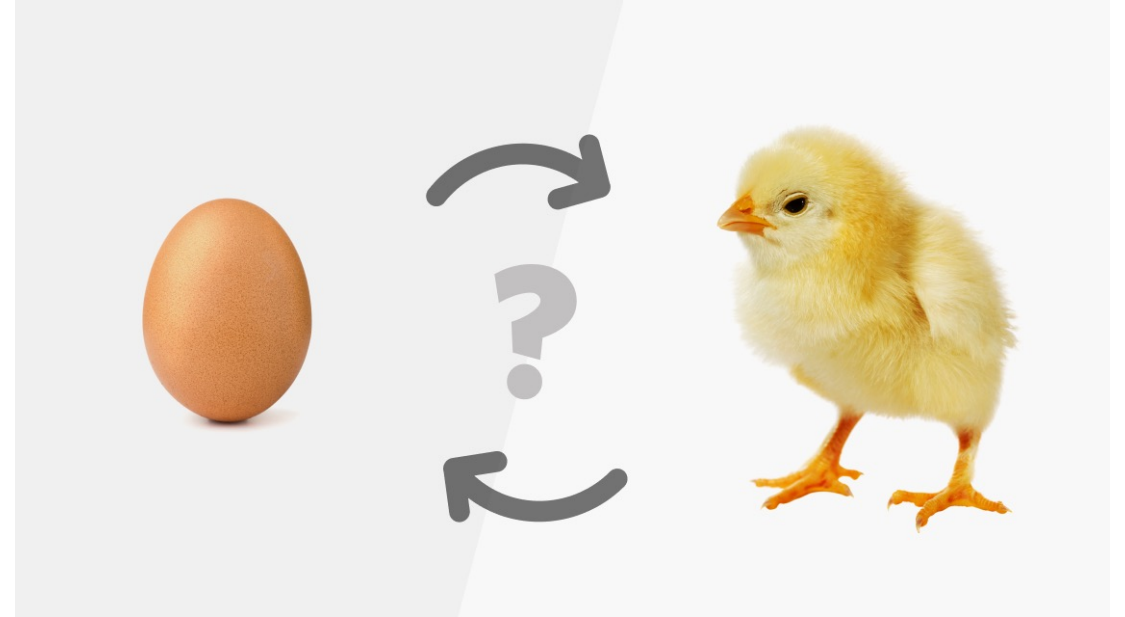
$$G_t \doteq R_{t+1} + \gamma R_{t+2} + \gamma^2 R_{t+3} + \cdots = \sum_{k=0}^{\infty} \gamma^k R_{t+k+1}$$

where γ is a discount factor, $\gamma \in [0,1]$ (0 = myopic, 1 = far-sighted)

- In RL we are not interested in maximizing the value of a single step, but we want to maximize the return (return = goal of RL).
- The discount is the **present value** of future rewards:
 - $\gamma = 0$ is the 'myopic' case (we give value only to present reward)
 - $\gamma = 1$ is the 'far-sighted' case (all rewards are important, even if far away in the future)

ii. Markov Reward Processes: Return

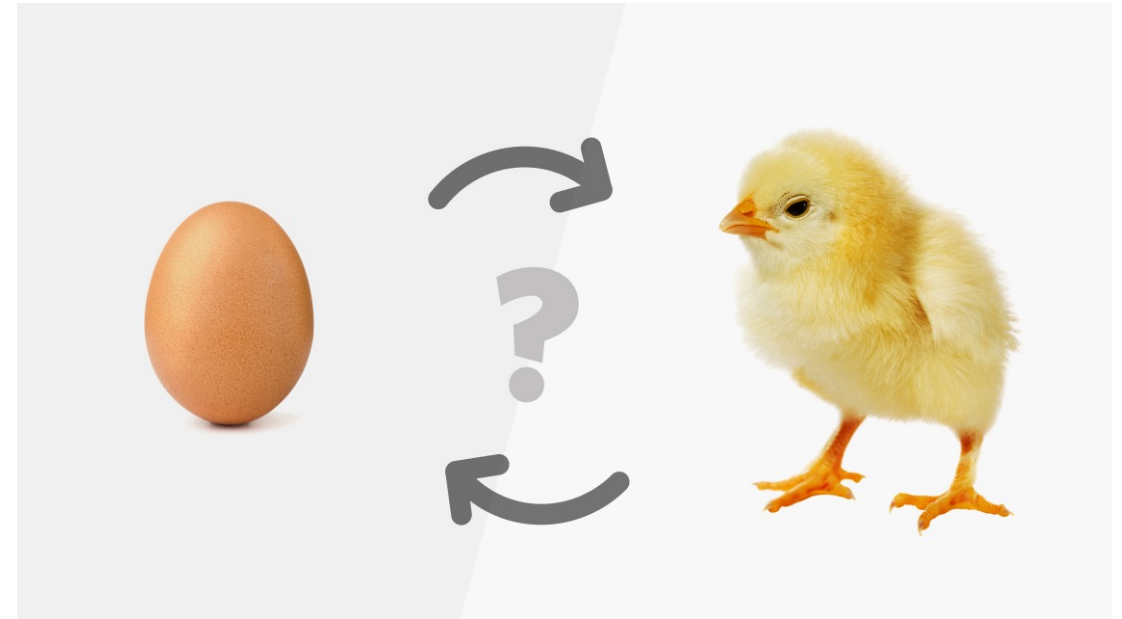
Why discounts?



ii. Markov Reward Processes: Return

Why discounts?

- Mathematically convenient: analysis can be simplified by the presence of rewards (for example we can avoid infinite returns in Markov Processes with cycles)
- Uncertainty about the future may not be fully represented
- Financial inspiration: immediate rewards may earn more interest than delayed rewards
- Animals and humans show preference for immediate rewards
- It is a general formulation: if $\gamma = 1$ we are considering the undiscounted Markov reward processes



ii. Markov Reward Processes: Value Function

The value function $v(s)$ gives the long-term value of state s

Definition

The **state value function** $v(s)$ of a Markov Reward Process is the expected return starting from state s

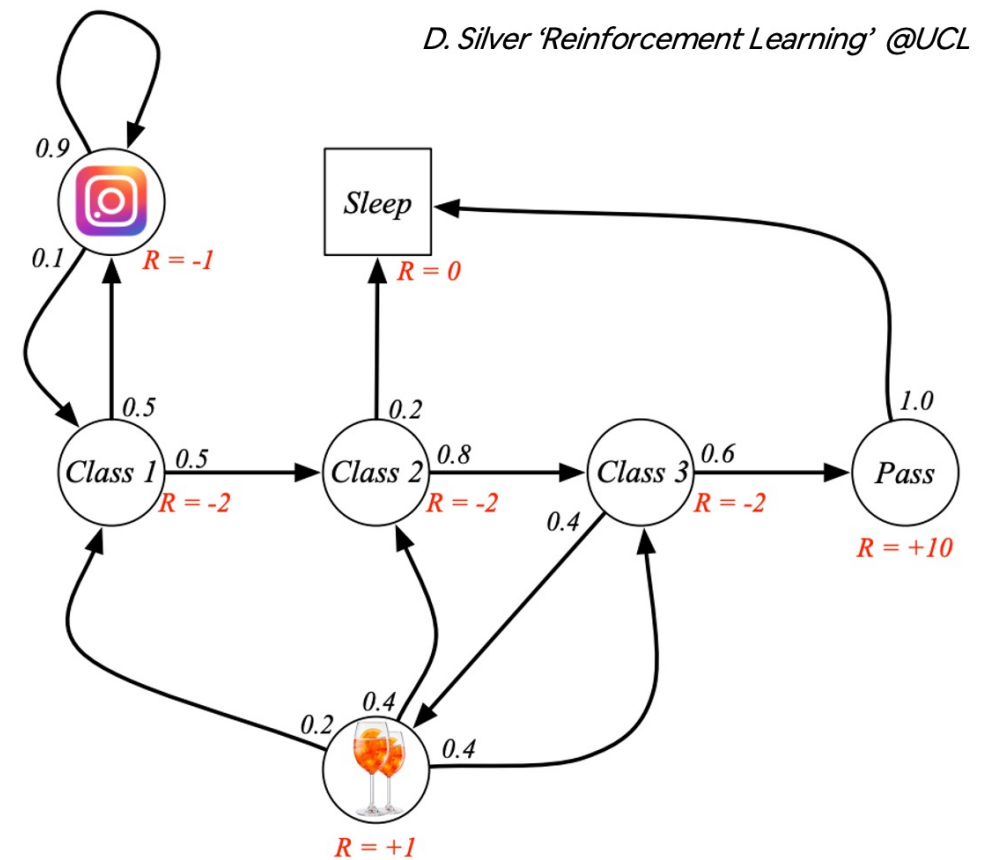
$$v(s) = \mathbb{E}[G_t | S_t = s]$$

Please note the expectation: this is fundamental since we are in stochastic settings

ii. Markov Reward Processes :

Student Markov Chain

Preview of Chapter 5 - How to compute state value functions from a Markov Reward Process?



ii. Markov Reward Processes :

Student Markov Chain

Let's consider $\gamma = 1/2$ and the return obtain with the available samples

C1 C2 C3 Pass Sleep -> $v(C1) = -2 - 2/2 - 2/4 + 10/8 = -2.25$

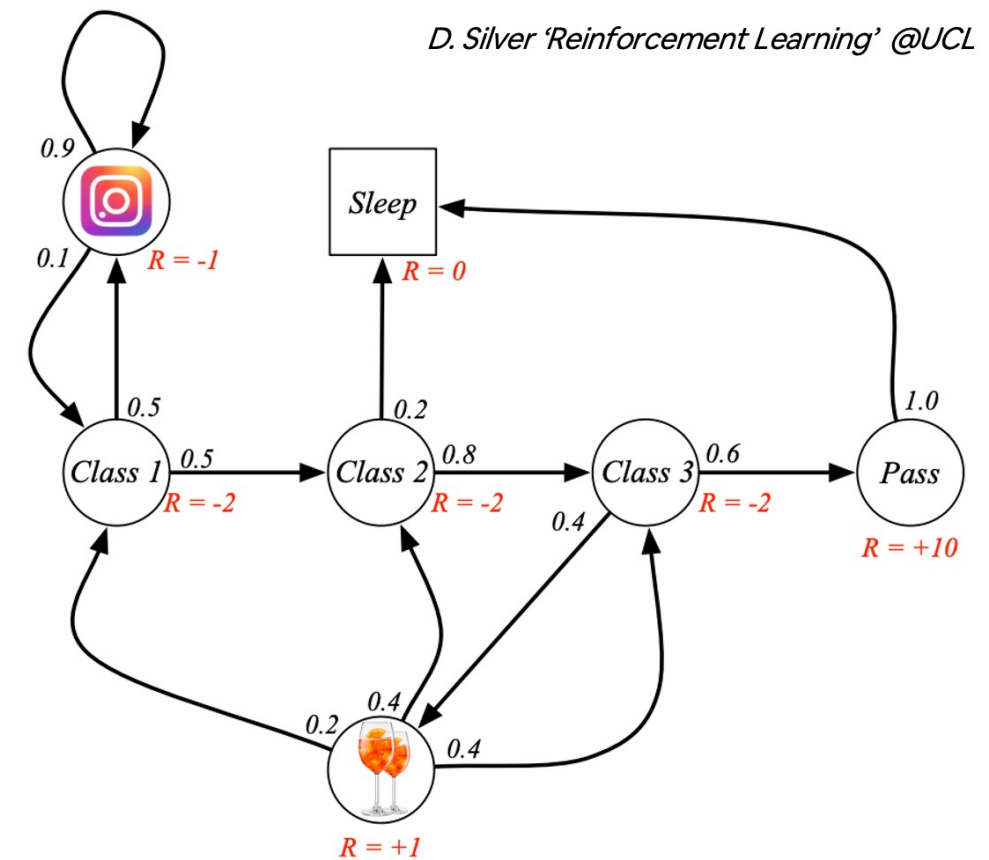
C1 IG IG C1 C2 Sleep -> $v(C1) = -2 - 1/2 - 1/4 - 2/8 - 2/16 = -3.125$

C1 C2 C3 Spritz C2 C3 Pass Sleep ->

$v(C1) = -2 - 2/2 - 2/4 + 1/8 - 2/16 - 2/32 + 10/64 = -3.41$

C1 IG IG C1 C2 C3 Spritz C1 IG IG IG C1 C2 C3 Spritz C2 Sleep

$v(C1) = -2 - 1/2 - 1/4 - 2/8 - 2/16 + \dots = -3.20$



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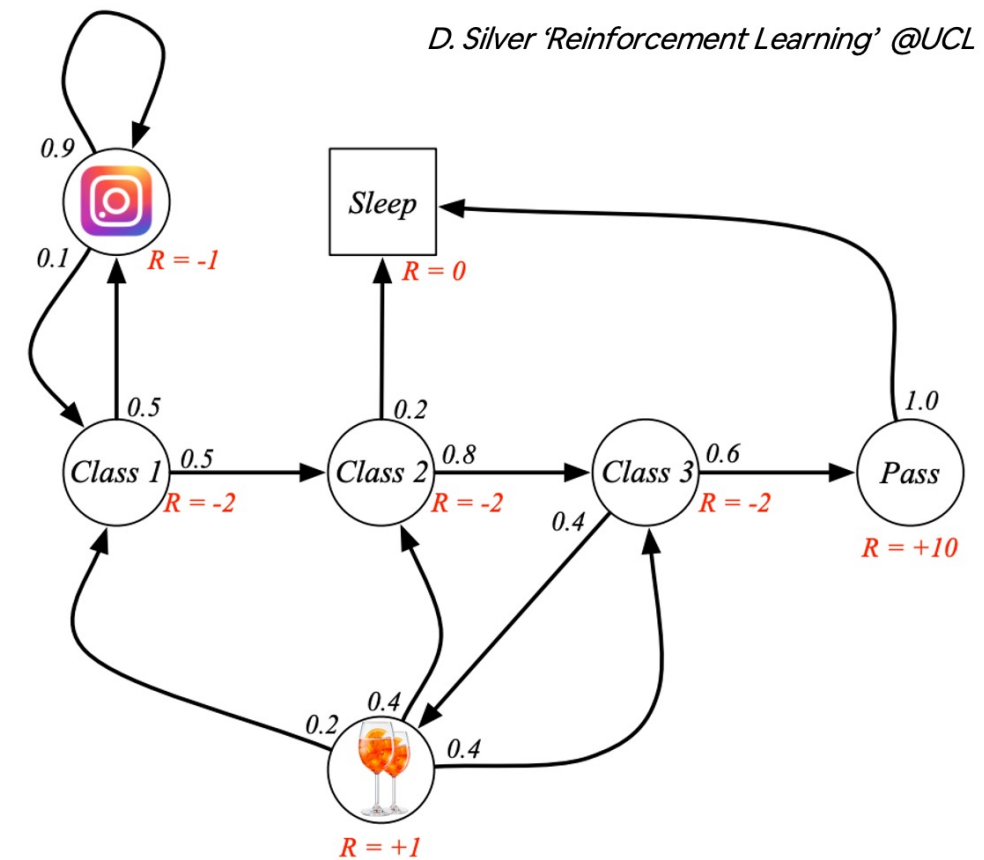
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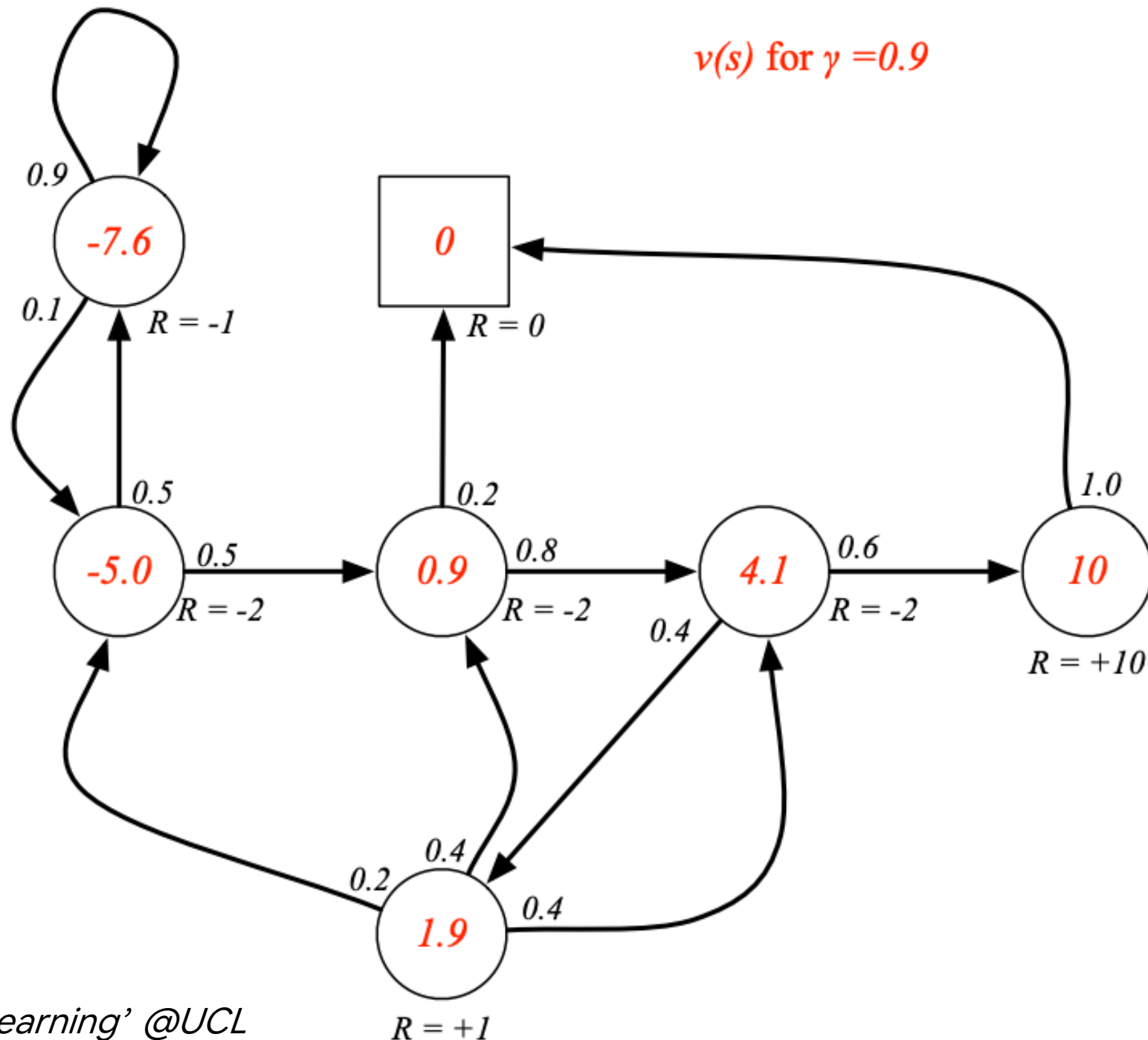


If we have samples, an estimate of the value function for state s is provided by the **sampled average of the returns seen from that state.**

I.e. In this case $v(C1) = (-2.25 - 3.125 - 3.41 - 3.20)/4 = -3$

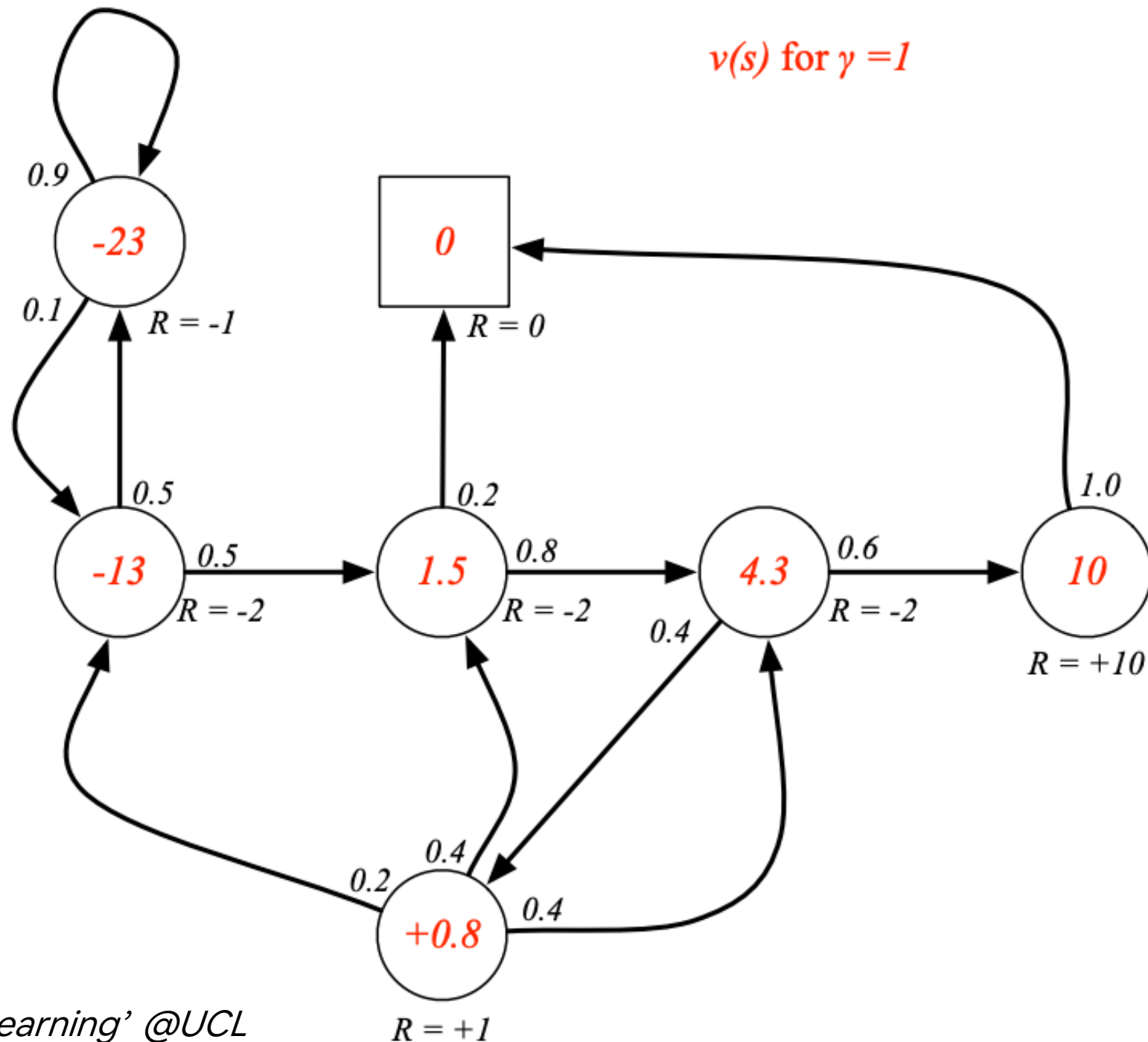
ii. Markov Reward Processes : Student Markov Chain

Different discounts values =
different state values
functions!



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The Bellman Equations

Bellman equations are our tool to 'solve' Markov Reward Processes (MRPs) and MDPs thanks to their recursive nature:

MRP	Bellman equation: for finding value functions	Linear: we can use it for 'small' MRPs. We need to resort to iterative approaches for 'large' MRPs
MDP	Bellman expectation equation: for finding value functions and action-value functions	Linear: we can use it for small MDPs. We need to resort to iterative approaches for 'large' MDPs
MDP	Bellman optimality equation: for finding optimal value functions and optimal action-value functions	Non-linear: we need iterative approaches even for small MDPs.

The Bellman Equations

Bellman equations are our tool to 'solve' Markov Reward Processes (MRPs) and MDPs thanks to their recursive nature:



Bellman equations are fundamental tools to understand v in MRPs and (v, q) in MDPs

When we will consider the 'true' RL problems, the algorithms that we will use, exploit Bellman equations

	Linear: we can use it for 'small' MRPs. We need to resort to iterative approaches for 'large' MRPs
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ii. Markov Reward Processes: Bellman Equation

The value function $v(s)$ can be decomposed into 2 parts:

- The immediate reward R_{t+1}
- The discounted value of successor state $\gamma v(S_{t+1})$

Exploiting

$$\begin{aligned} G_t &\doteq R_{t+1} + \gamma R_{t+2} + \gamma^2 R_{t+3} + \gamma^3 R_{t+4} + \cdots \\ &= R_{t+1} + \gamma(R_{t+2} + \gamma R_{t+3} + \gamma^2 R_{t+4} + \cdots) \\ &= R_{t+1} + \gamma G_{t+1} \end{aligned}$$

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We have that (law of iterated expectations)

$$\begin{aligned} v(s) &= \mathbb{E}[G_t | S_t = s] = \mathbb{E}[R_{t+1} + \gamma G_{t+1} | S_t = s] \\ &= \mathbb{E}[R_{t+1} + \gamma v(S_{t+1}) | S_t = s] \end{aligned}$$

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The value of a state is in relationship with the next one!

We will use 'iterative definitions' many times throughout the course!

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Pay Attention: we will have 3 different 'Bellman Equations' today

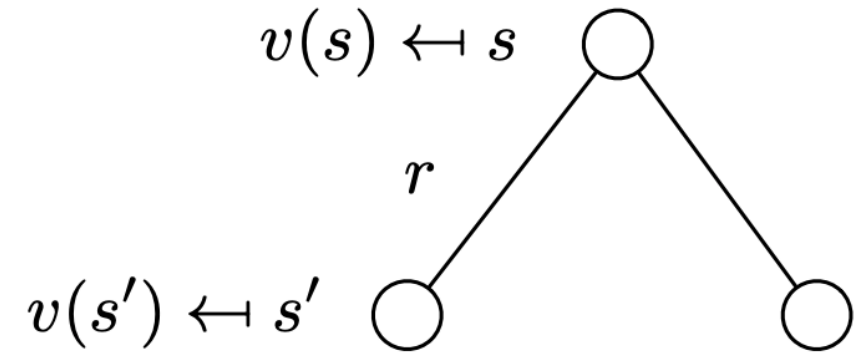
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ii. Markov Reward Processes: Bellman Equation

The definition of the value-function from the Bellman Equation can be seen a **1-step look ahead search**

Backup diagrams: visual representations of different algorithms and models in RL



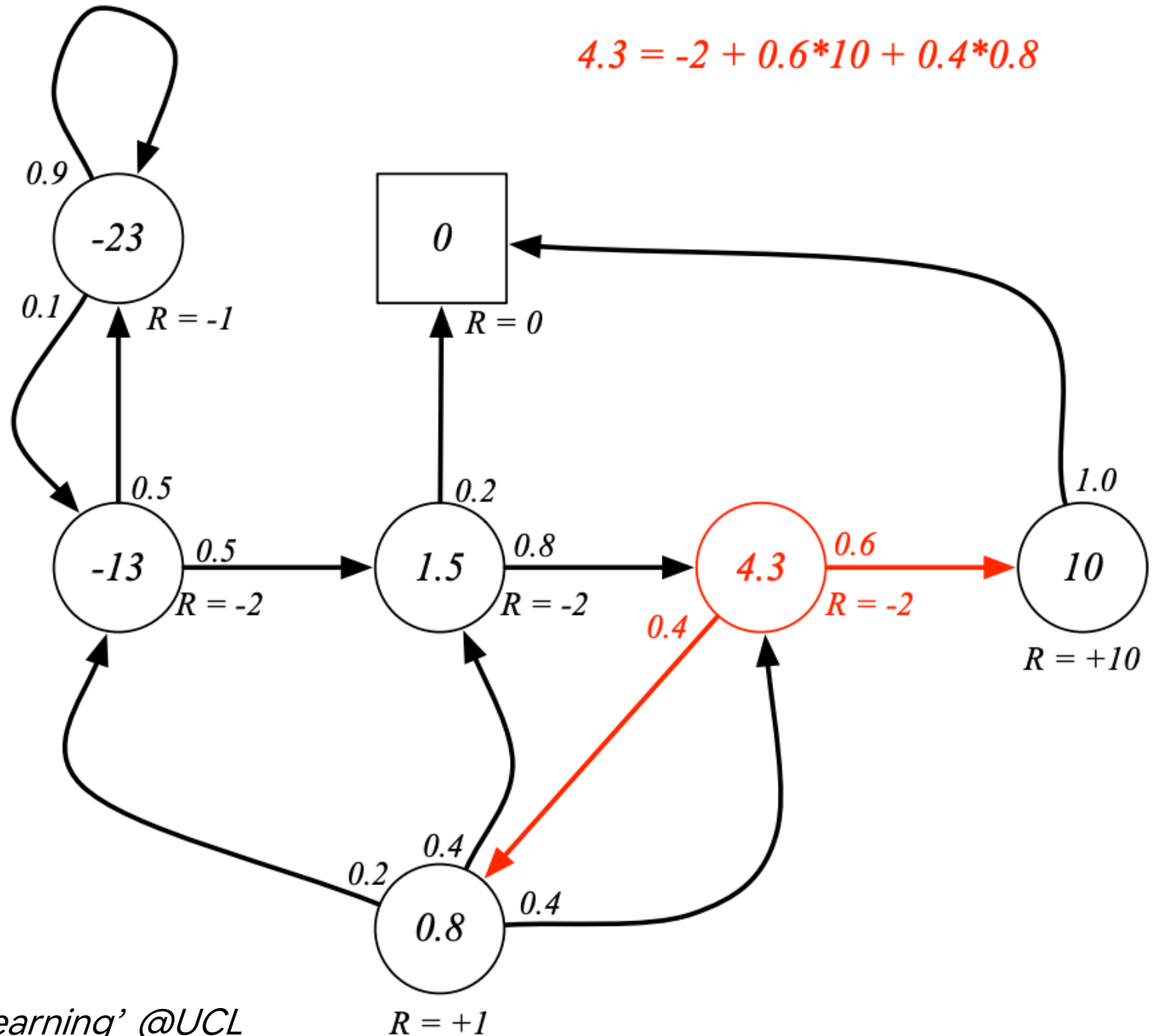
$$v(s) = \mathbb{E} [R_{t+1} + \gamma v(S_{t+1}) \mid S_t = s]$$

We need to consider all possible successor states with the related transition probabilities

$$v(s) = \mathcal{R}_s + \gamma \sum_{s' \in \mathcal{S}} \mathcal{P}_{ss'} v(s')$$

ii. Markov Reward Processes : Student Markov Chain

- Undiscounted case ($\gamma = 1$)
- If state value functions are provided we can use the Bellman Equation to verify if they are true



ii. Markov Reward Processes: Bellman Equation in Matrix Form

The Bellman Equation can be written concisely in matrix form

$$\boldsymbol{v} = \boldsymbol{\mathcal{R}} + \gamma \boldsymbol{\mathcal{P}} \boldsymbol{v}$$

Where $\boldsymbol{v}$ is a column vector with one entry per state

$$\begin{bmatrix} v(1) \\ \vdots \\ v(n) \end{bmatrix} = \begin{bmatrix} \mathcal{R}_1 \\ \vdots \\ \mathcal{R}_n \end{bmatrix} + \gamma \begin{bmatrix} \mathcal{P}_{11} & \dots & \mathcal{P}_{1n} \\ \vdots & & \\ \mathcal{P}_{n1} & \dots & \mathcal{P}_{nn} \end{bmatrix} \begin{bmatrix} v(1) \\ \vdots \\ v(n) \end{bmatrix}$$

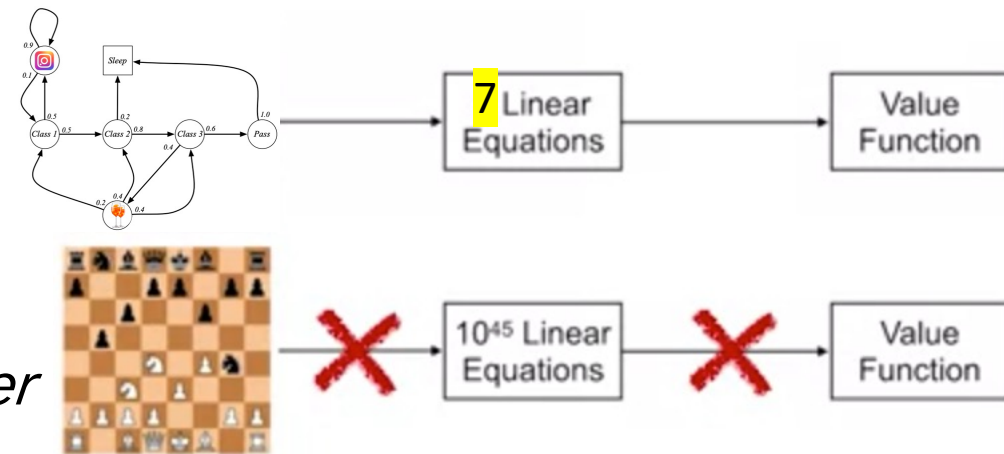
ii. Markov Reward Processes: Solving the Bellman Equation

- The Bellman Equation (when we are dealing with **Markov Reward Processes***) is linear
- We can solve the previous matrix directly
- If we have n states, the computational cost is $O[n^3]$: affordable only with small Markov Reward Processes (MRPs)
- For large MRPs we will look for efficient methods (Dynamic Programming, Monte-Carlo evaluation, Temporal-Difference learning)

$$v = \mathcal{R} + \gamma \mathcal{P}v$$

$$(I - \gamma \mathcal{P})v = \mathcal{R}$$

$$v = (I - \gamma \mathcal{P})^{-1} \mathcal{R}$$



This will not be true when we will deal with optimization and maximization (when we will consider an agent making decisions!) in **Markov Decision Processes*

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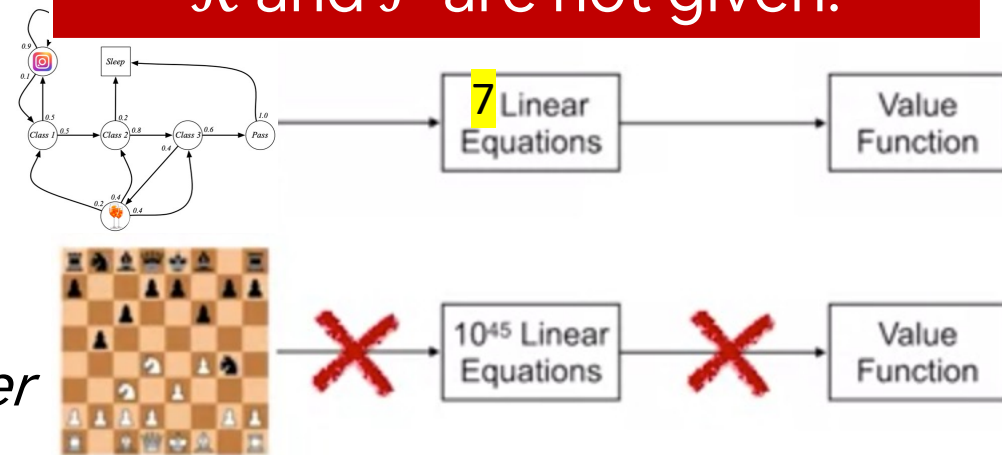
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$$v = \mathcal{R} + \gamma \mathcal{P} v$$

$$(I - \gamma \mathcal{P}) v = \mathcal{R}$$

$$v = (I - \gamma \mathcal{P})^{-1} \mathcal{R}$$

Moreover, in true RL problems $\mathcal{R}$ and $\mathcal{P}$ are not given!



This will not be true when we will deal with optimization and maximization (when we will consider an agent making decisions!) in **Markov Decision Processes*

iii. Markov Decision Processes (MDPs)

Let's add actions and decisions (a true RL problem!): a Markov Decision Process is a Reward Process with decisions. MDP is an environment in which all states are Markov.

Definition

A **Markov Decision Process** is a tuple $\langle \mathcal{S}, \mathcal{A}, \mathcal{P}, \mathcal{R}, \gamma \rangle$ such that:

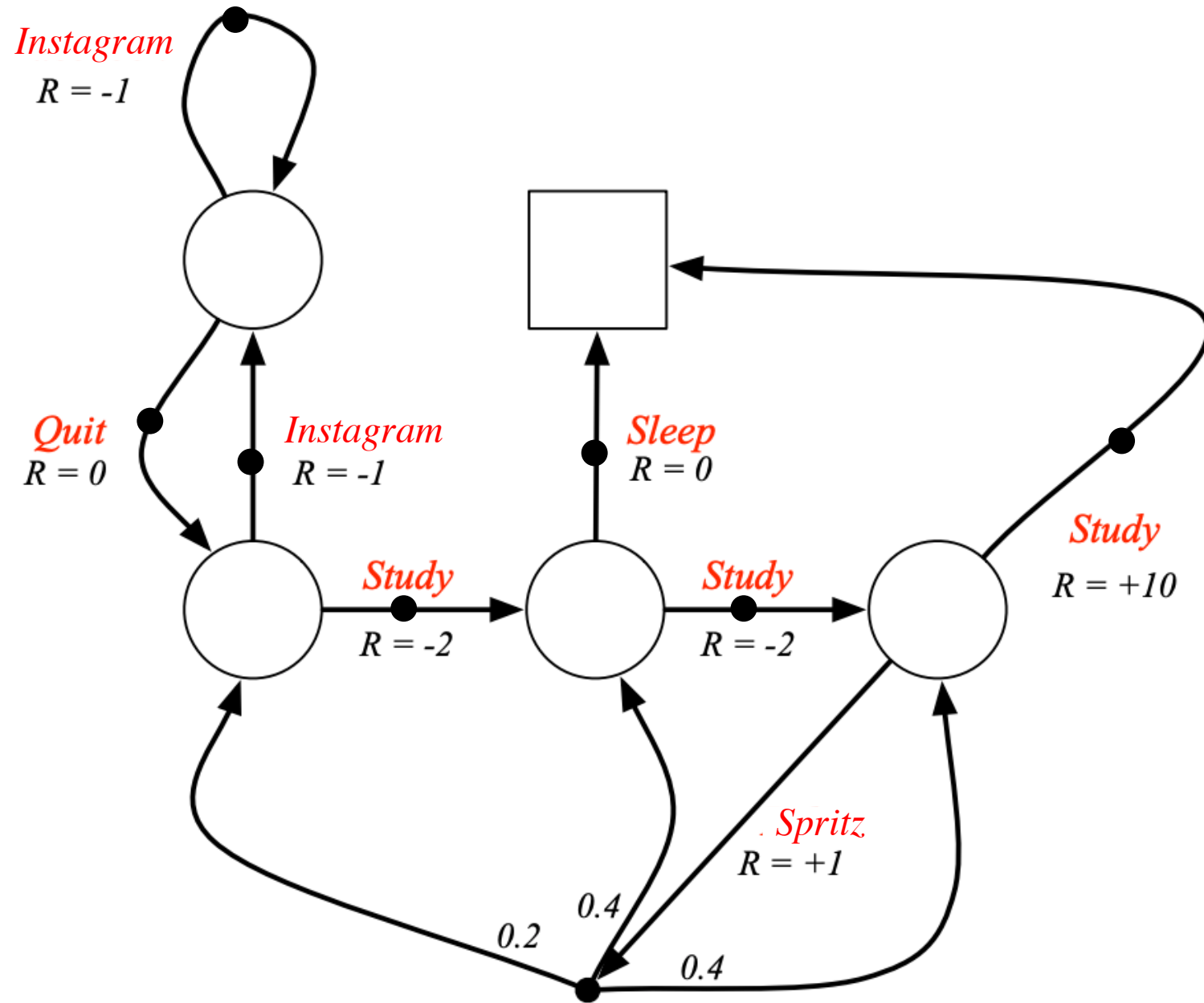
- $\mathcal{S}$ is a finite set of states
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- $\mathcal{P}$ is a state transition probability matrix with entries

$$\mathcal{P}_{ss'}^a = \mathbb{P}[S_{t+1} = s' | S_t = s, A_t = a]$$

- $\mathcal{R}$ is a reward function, $\mathcal{R}_s^a = \mathbb{E}[R_{t+1} | S_t = s, A_t = a]$ (it is just the immediate reward, in that specific state)
- γ is a discount factor, $\gamma \in [0, 1]$

iii. Markov Decision Processes: Student Markov Chain

- Pay attention: we are reporting **actions** (typically indicated with a black dot)
- States are different here
- Now there is control and agency! We should define a policy!
- Now we can try to maximize our reward!



iii. MDPs: (Stochastic) Policies

Definition

A **Policy** π is a distribution over actions given that we are in a state:

$$\pi(a|s) = \mathbb{P}[A_t = a | S_t = s]$$

- A **policy fully defines the behaviour of an agent**
- MDP policies depend on the current state (we are considering Markov states in MDPs, history doesn't matter)
- In **MDP policies are stationary** (do not depend on t), however we can change our policy in future episodes
- We consider **stochastic policies**: this allow us for example to deal with exploration!
- Please note that there is no reward here: the policy can be given or we may have 'learned' the policy with a dedicated procedure

iii. MDPs: (Stochastic) Policies

Given an MDP $\langle \mathcal{S}, \mathcal{A}, \mathcal{P}, \mathcal{R}, \gamma \rangle$ + a policy π :

- The state sequence $S_1, S_2, \dots$ is a Markov Process $\langle \mathcal{S}, \mathcal{P}^\pi \rangle$ where

$$\mathcal{P}_{s,s'}^\pi = \sum_{a \in \mathcal{A}} \pi(a|s) \mathcal{P}_{s,s'}^a$$

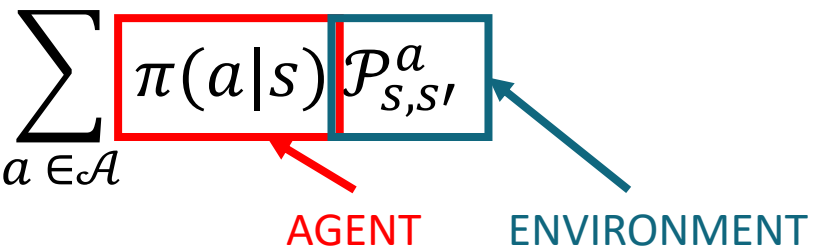
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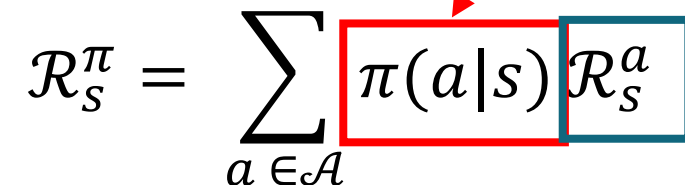
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iii. MDPs: Value Function

We had value functions with Markov Reward Processes, but now that we have agency, value of a state depends on the policy!

Definitions

The **state-value function** $v_{\pi}(s)$ of an MDP is the expected return from state s if we follow policy π :

$$v_{\pi}(s) = \mathbb{E}_{\pi}[G_t | S_t = s]$$

The **action-value function** $q_{\pi}(s, a)$ of an MDP is the expected return from state s if we take action a and then we follow policy π :

$$q_{\pi}(s, a) = \mathbb{E}_{\pi}[G_t | S_t = s, A_t = a]$$

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This is defined for all actions, also for the ones that are taken by π in state s

iii. MDPs: Value Function – Why it depends on π



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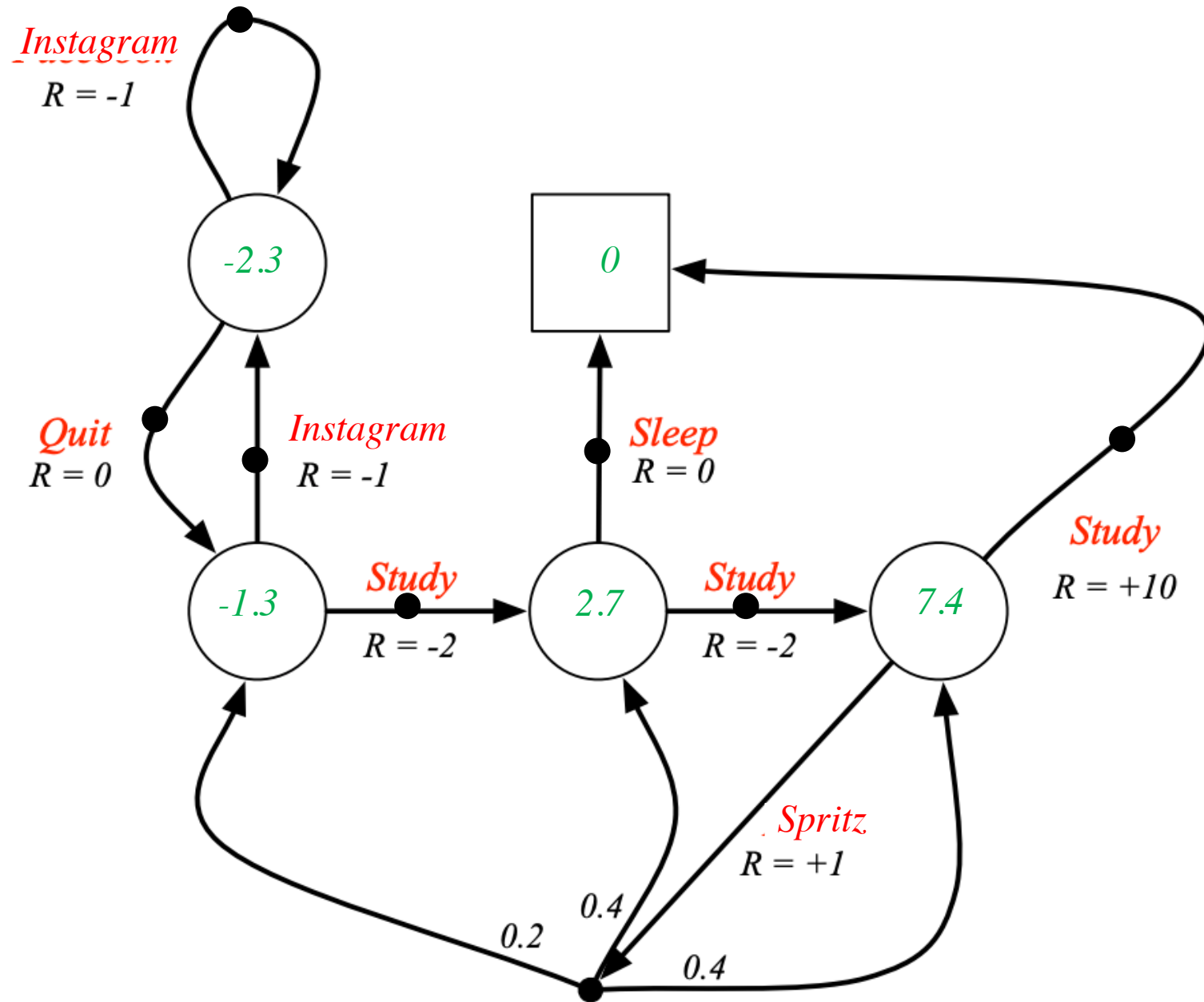
$$V_{Magnus} \neq V_{GianAntonio}$$



iii. Markov Decision Processes: Student Markov Chain

We consider the undiscounted MDP ($\gamma = 1$) and a uniform random policy: for each state (C1, IG, C2, C3) there are two possible actions, each one with probability 0.5

$$\pi(a|s) = 0.5 \text{ for all } a, s$$



iii. MDPs: Bellman Expectation Equation

The value function $v_\pi(s)$ can again be decomposed into 2 parts:

- The immediate reward R_{t+1}
- The discounted value of successor state $\gamma v_\pi(S_{t+1})$

$$v_\pi(s) = \mathbb{E}[R_{t+1} + \gamma v_\pi(S_{t+1}) | S_t = s]$$

Similarly for the action-value function

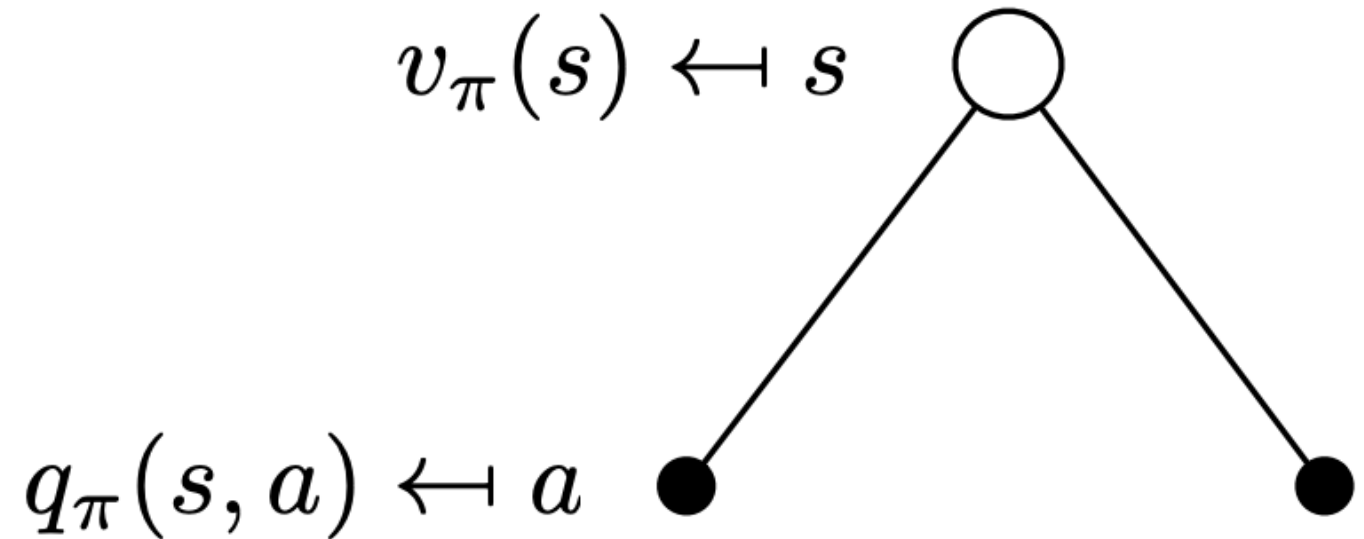
$$q_\pi(s, a) = \mathbb{E}[R_{t+1} + \gamma q_\pi(S_{t+1}, A_{t+1}) | S_t = s, A_t = a]$$

* Please note that previously(for the Bellman Equation) we have considered $v(s)$

iii. MDPs: Bellman Expectation Equation

Let's see the relation
between q_π and v_π (1 of 4)

states: ○
actions: ●



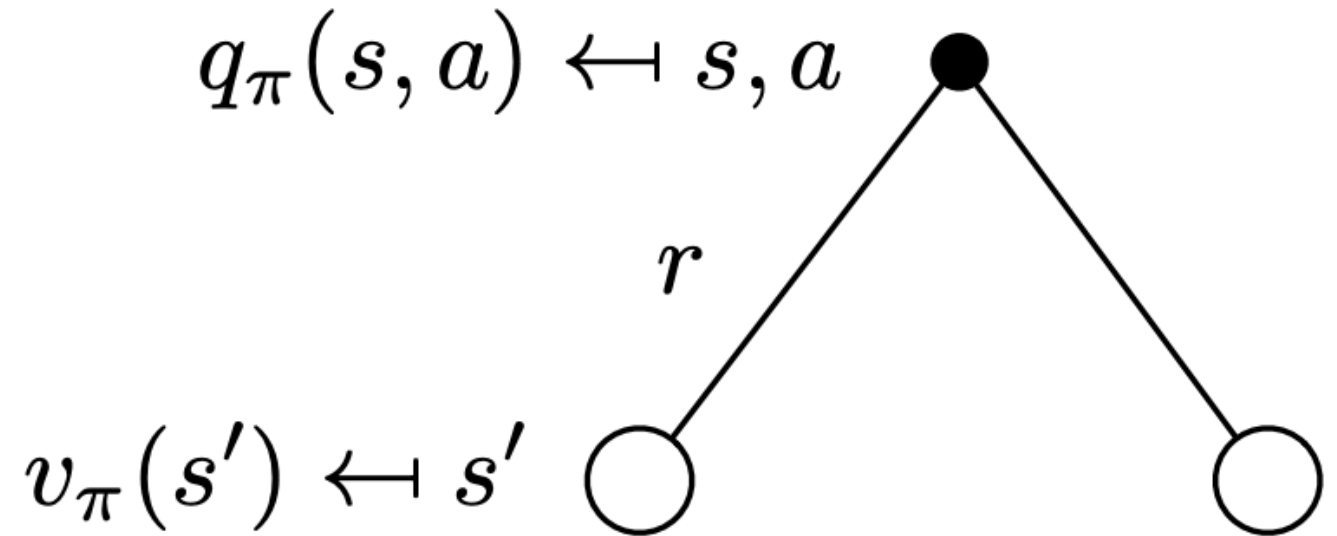
$$v_\pi(s) = \mathbb{E}_\pi[G_t | S_t = s]$$
$$q_\pi(s, a) = \mathbb{E}_\pi[G_t | S_t = s, A_t = a]$$

$$v_\pi(s) = \sum_{a \in \mathcal{A}} \pi(a|s) q_\pi(s, a)$$

iii. MDPs: Bellman Expectation Equation

Let's see the relation
between q_π and v_π (2 of 4)

states: ○
actions: ●



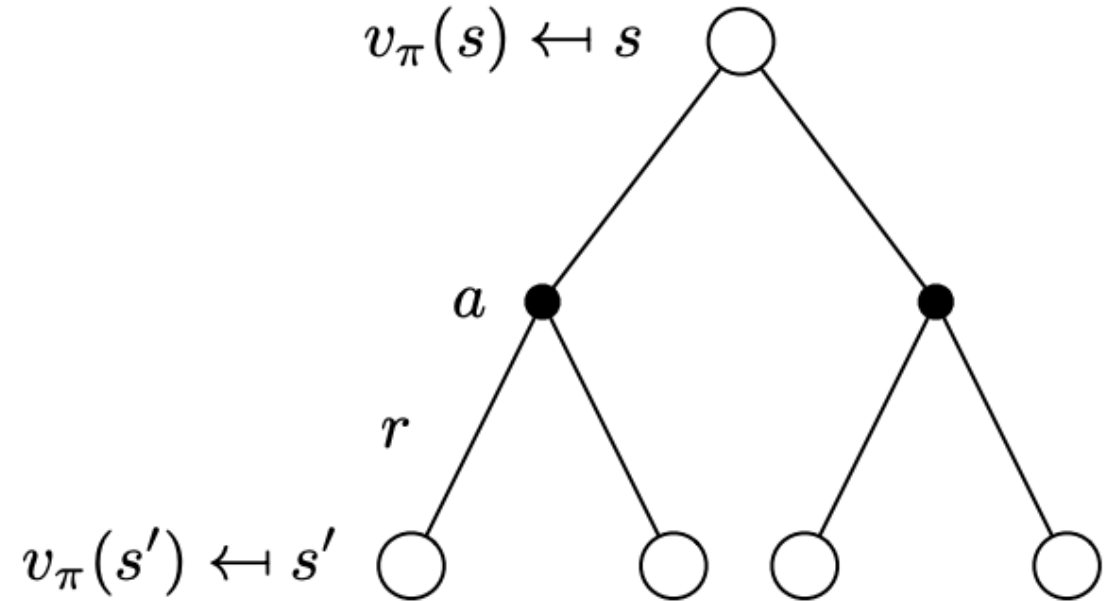
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iii. MDPs: Bellman Expectation Equation

Let's see the relation
between q_π and v_π (3 of 4)

states: ○
actions: ●



$$v_\pi(s) = \sum_{a \in \mathcal{A}} \pi(a|s) q_\pi(s, a)$$

$$q_\pi(s, a) = \mathcal{R}_s^a + \gamma \sum_{s' \in \mathcal{S}} \mathcal{P}_{ss'}^a v_\pi(s')$$

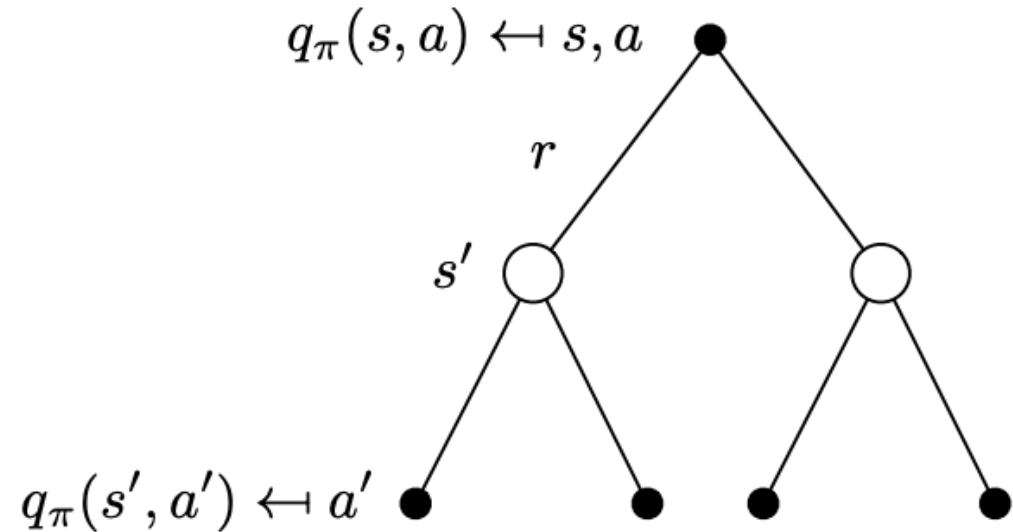
We can obtain a recursive
description of v_π

$$v_\pi(s) = \sum_{a \in \mathcal{A}} \pi(a|s) \left(\mathcal{R}_s^a + \gamma \sum_{s' \in \mathcal{S}} \mathcal{P}_{ss'}^a v_\pi(s') \right)$$

iii. MDPs: Bellman Expectation Equation

Let's see the relation
between q_π and v_π (4 of 4)

states: ○
actions: ●



We obtain a recursive
description of q_π

$$v_\pi(s) = \sum_{a \in \mathcal{A}} \pi(a|s) q_\pi(s, a)$$

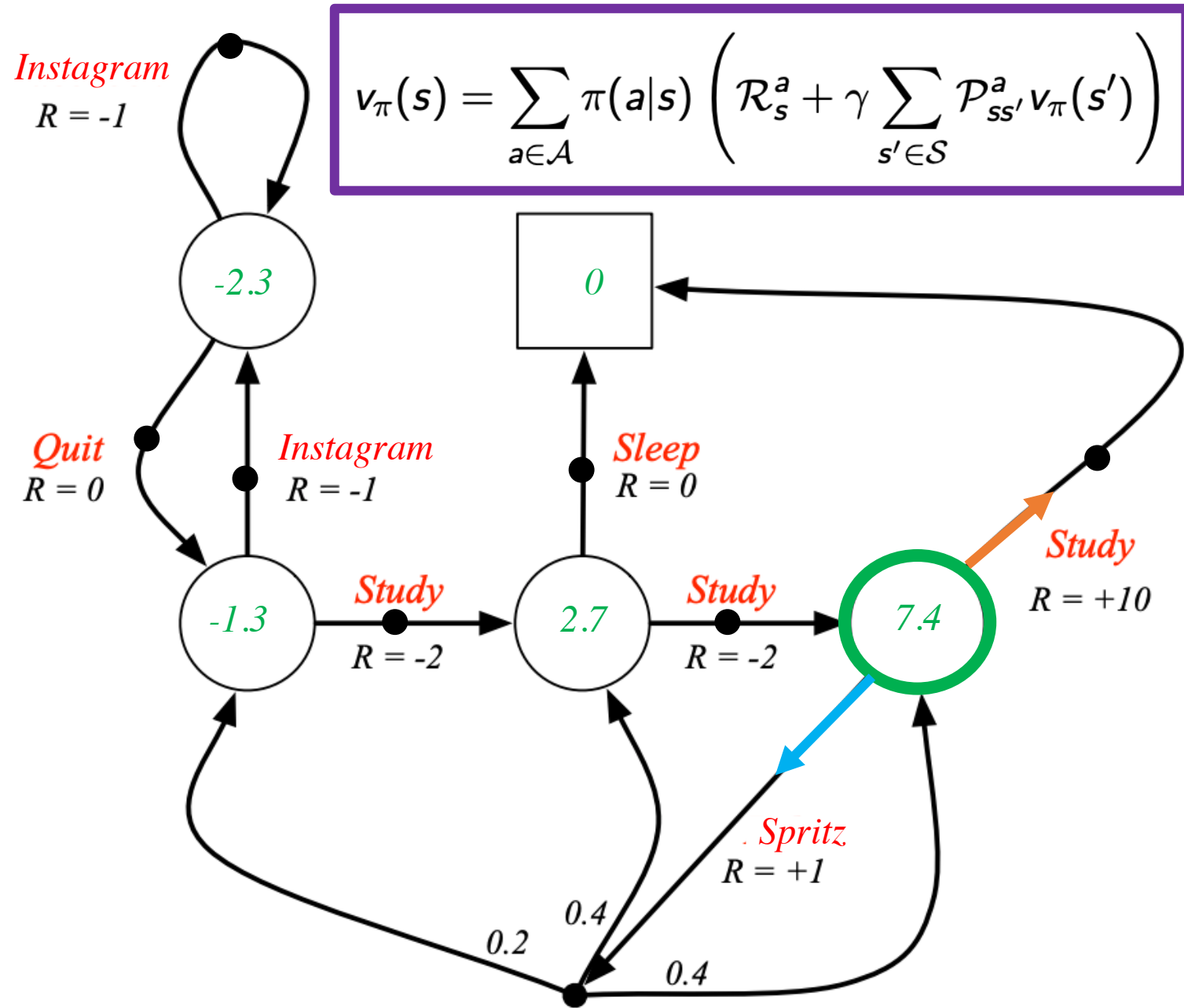
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$$q_\pi(s, a) = \mathcal{R}_s^a + \gamma \sum_{s' \in \mathcal{S}} \mathcal{P}_{ss'}^a \sum_{a' \in \mathcal{A}} \pi(a'|s') q_\pi(s', a')$$

iii. Markov Decision Processes: Student Markov Chain

Let's consider the previous case (undiscounted, uniform random policy) and let's verify with the recursive definition that $v_{\pi}(C3) = 7.4$

$$7.4 = 0.5 \cdot 10 + 0.5 \cdot (1 + 0.4 \cdot 7.4 - 0.2 \cdot 1.3 + 0.4 \cdot 2.7)$$



iii. Markov Decision Processes: Bellman Expectation Equation in Matrix Form

Also the Bellman Expectation Equation can be written concisely in matrix form (we are resorting to the induced Markov Reward Process by using $\mathcal{R}_s^\pi = \sum_{a \in \mathcal{A}} \pi(a|s) \mathcal{R}_s^a$ seen before):

$$v_\pi = \mathcal{R}^\pi + \gamma \mathcal{P}^\pi v_\pi$$

That can be solved:

$$v_\pi = (I - \gamma \mathcal{P}^\pi)^{-1} \mathcal{R}^\pi$$

As said, a good part of the course will be dedicated to find efficient ways to avoid computing such set of linear equations.

Optimal Value Function

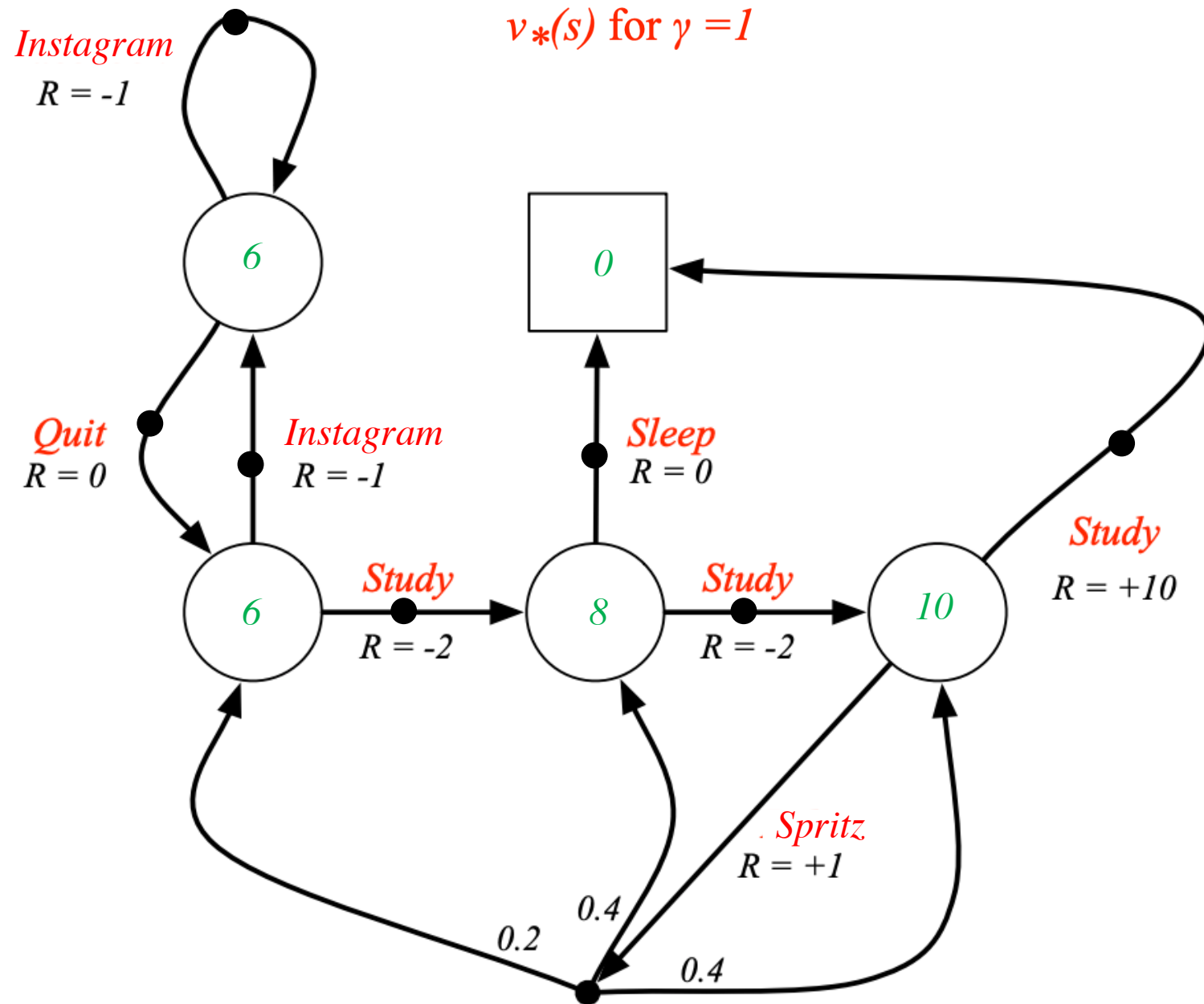
Definition

The **optimal state-value function** $v_*(s)$ is the maximum state-value function over all policies:

$$v_*(s) = \max_{\pi} v_{\pi}(s)$$

Optimal **value** function on the Student Markov Chain

- Since here we have the deterministic case (one action from one state lead you to a deterministic successor state) we can easily compute the value functions for each state going backwards
- We still don't have an explicit indication on how to behave



Optimal Value Function

Definition

The **optimal state-value function** $v_*(s)$ is the maximum state-value function over all policies:

$$v_*(s) = \max_{\pi} v_{\pi}(s)$$

The **optimal action-value function** $q_*(s)$ is the maximum action-value function over all policies:

$$q_*(s, a) = \max_{\pi} q_{\pi}(s, a)$$

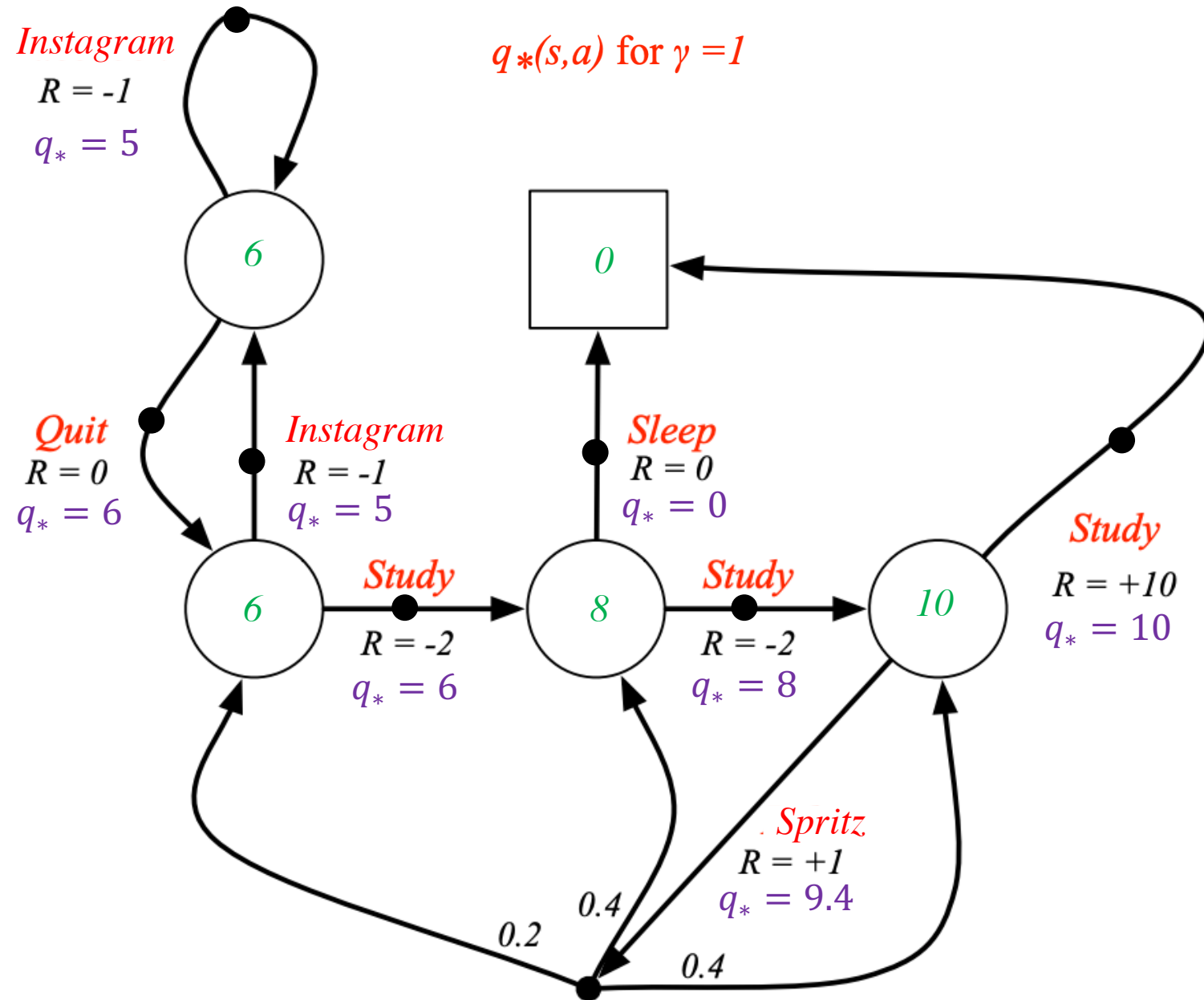
An MDP can be considered ‘solved’ on a RL perspective when the optimal action-value function is known: we always know the optimal action to take from a certain state

Optimal **action** value function on the Student Markov Chain

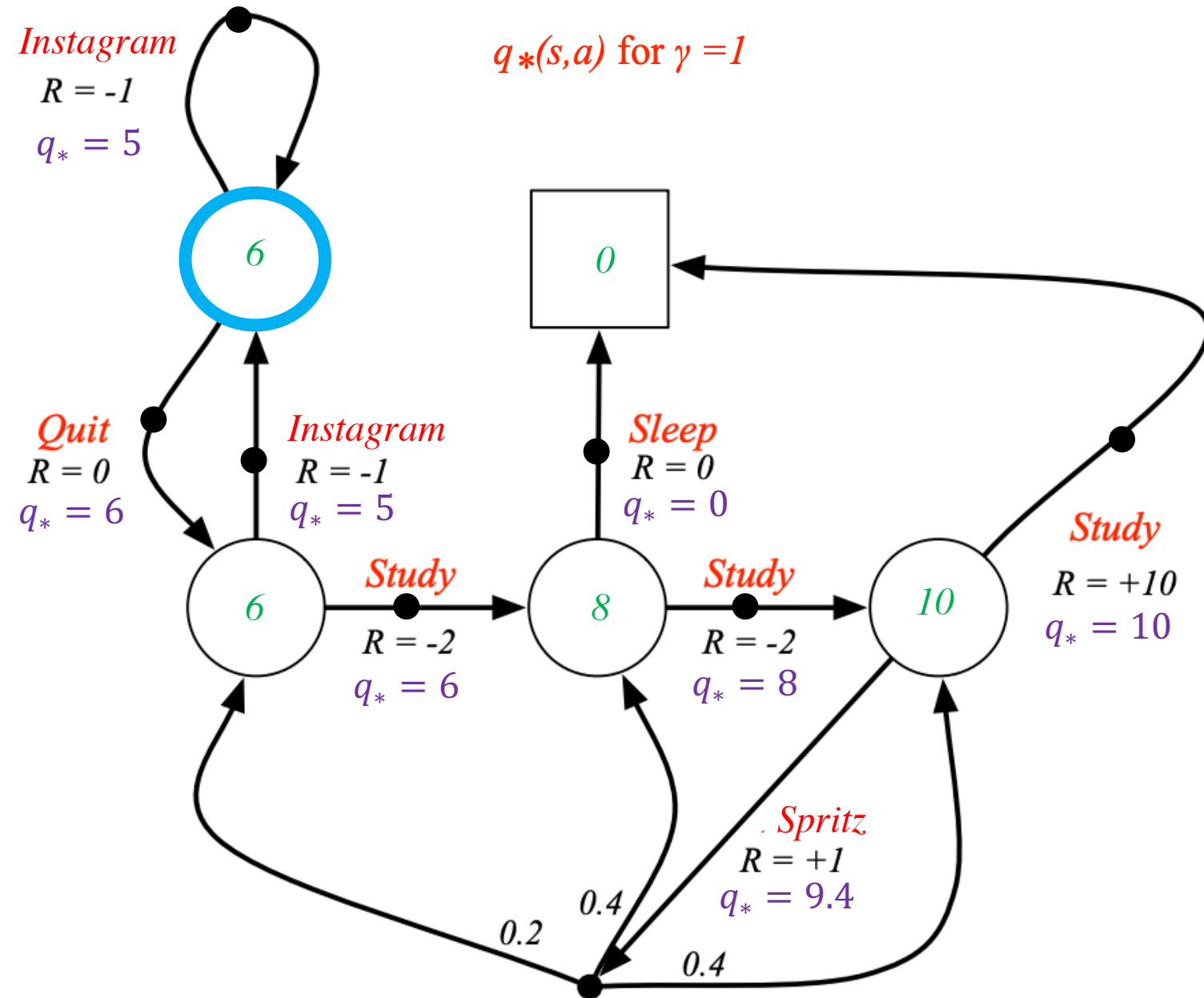
The following holds also for the optimal policy:

$$q_*(s, a) = \max_{\pi} q_{\pi}(s, a)$$

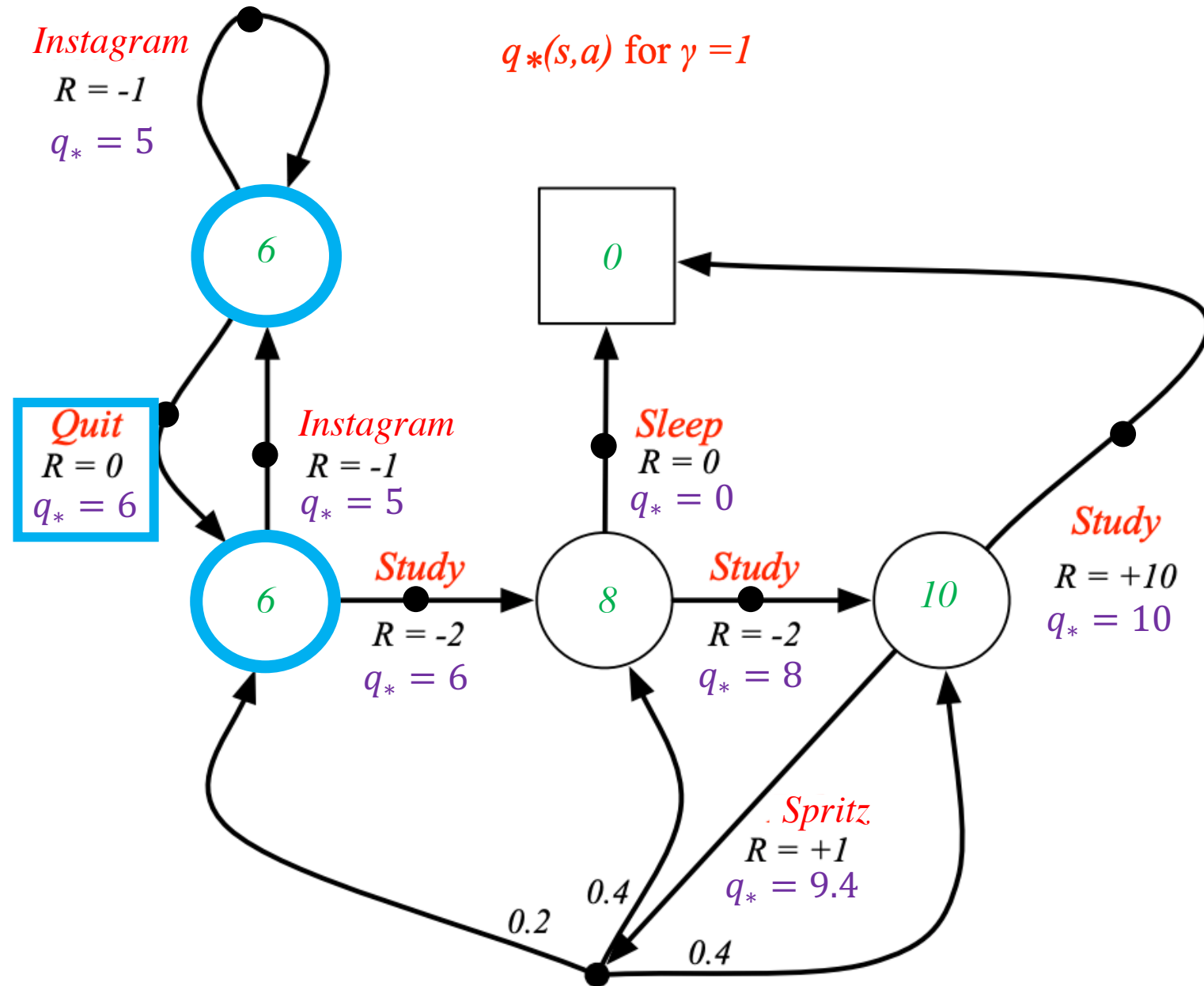
Action value function allows us to take decisions...



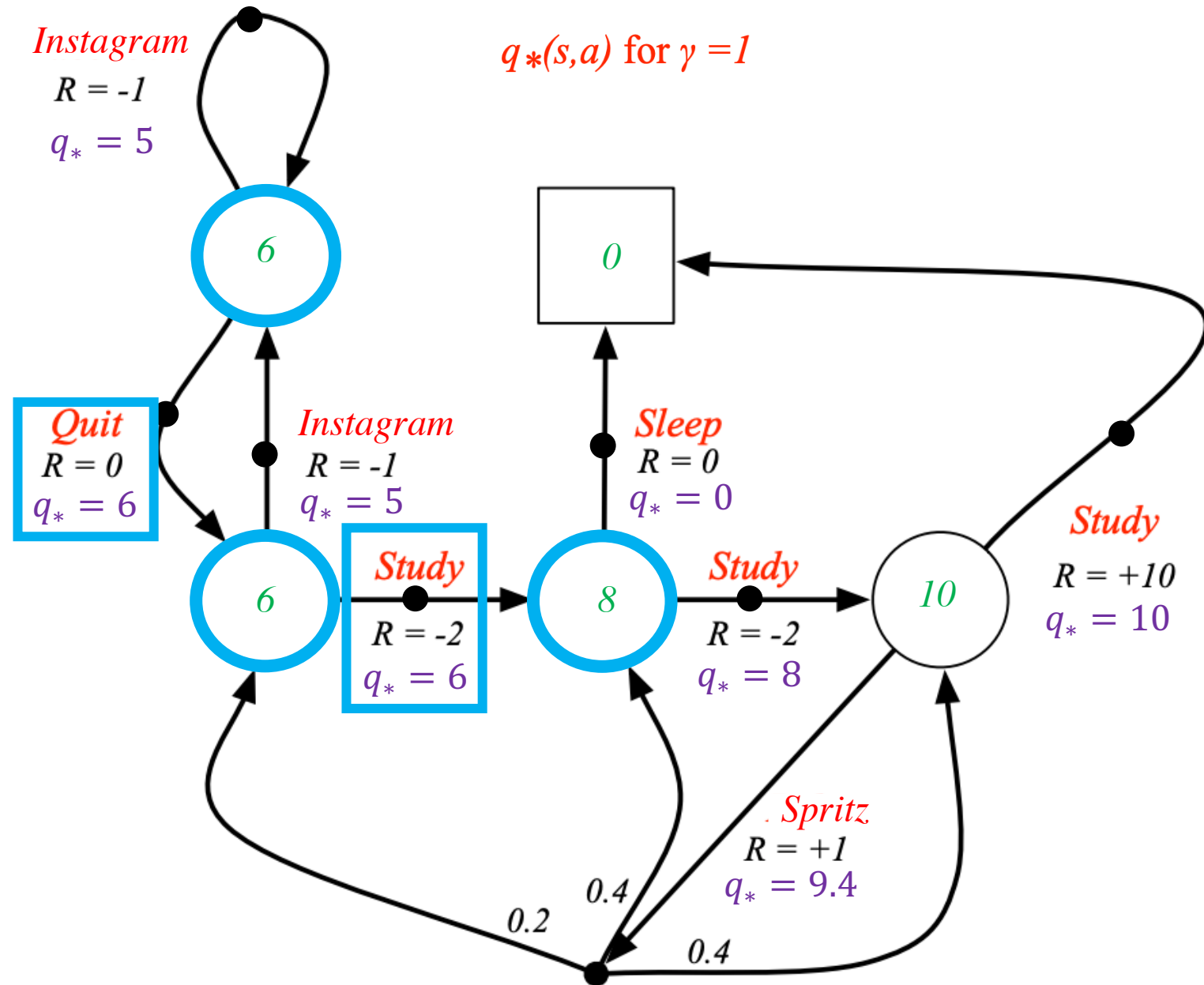
Optimal **action** value function on the Student Markov Chain



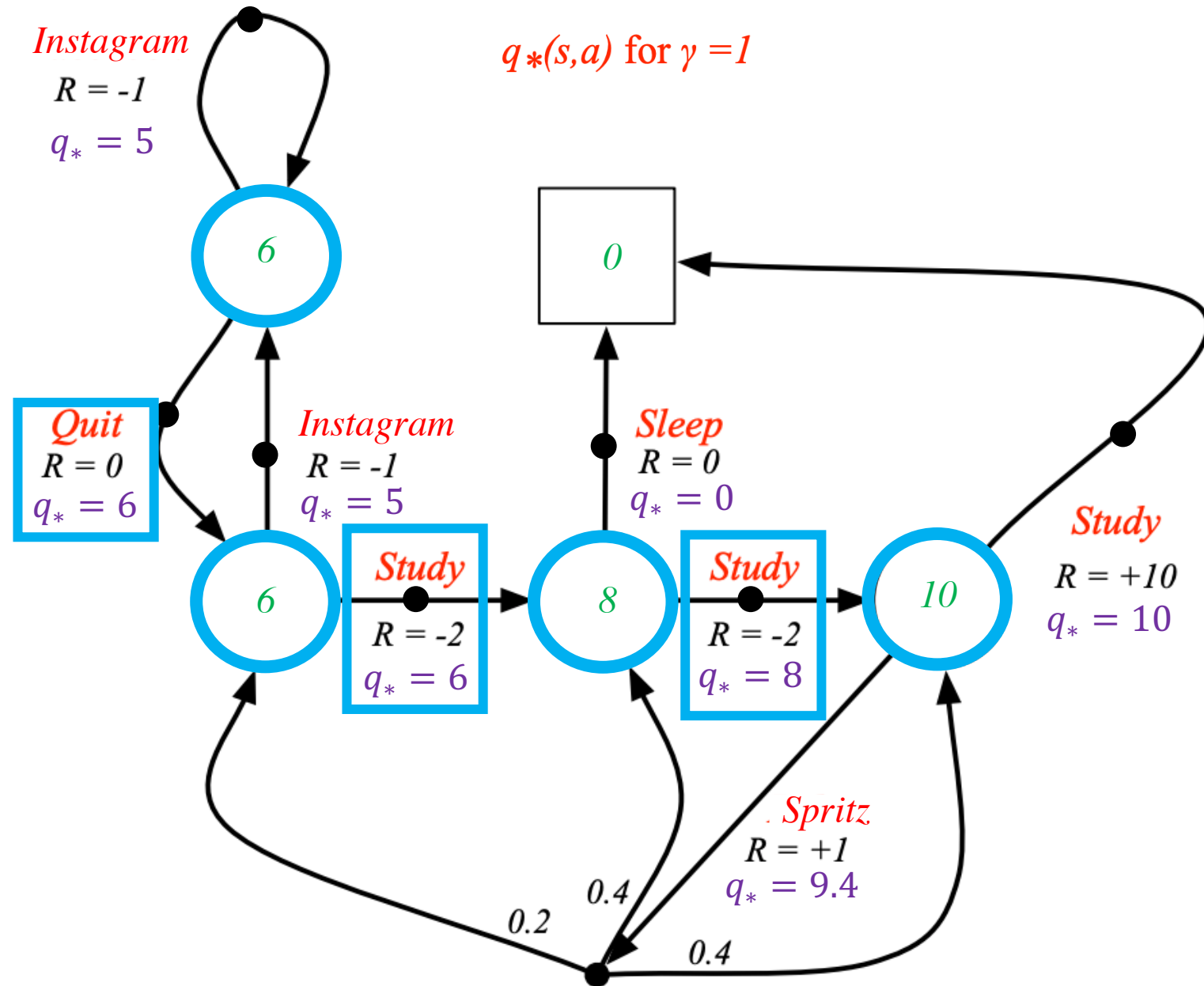
Optimal **action** value function on the Student Markov Chain



Optimal **action** value function on the Student Markov Chain



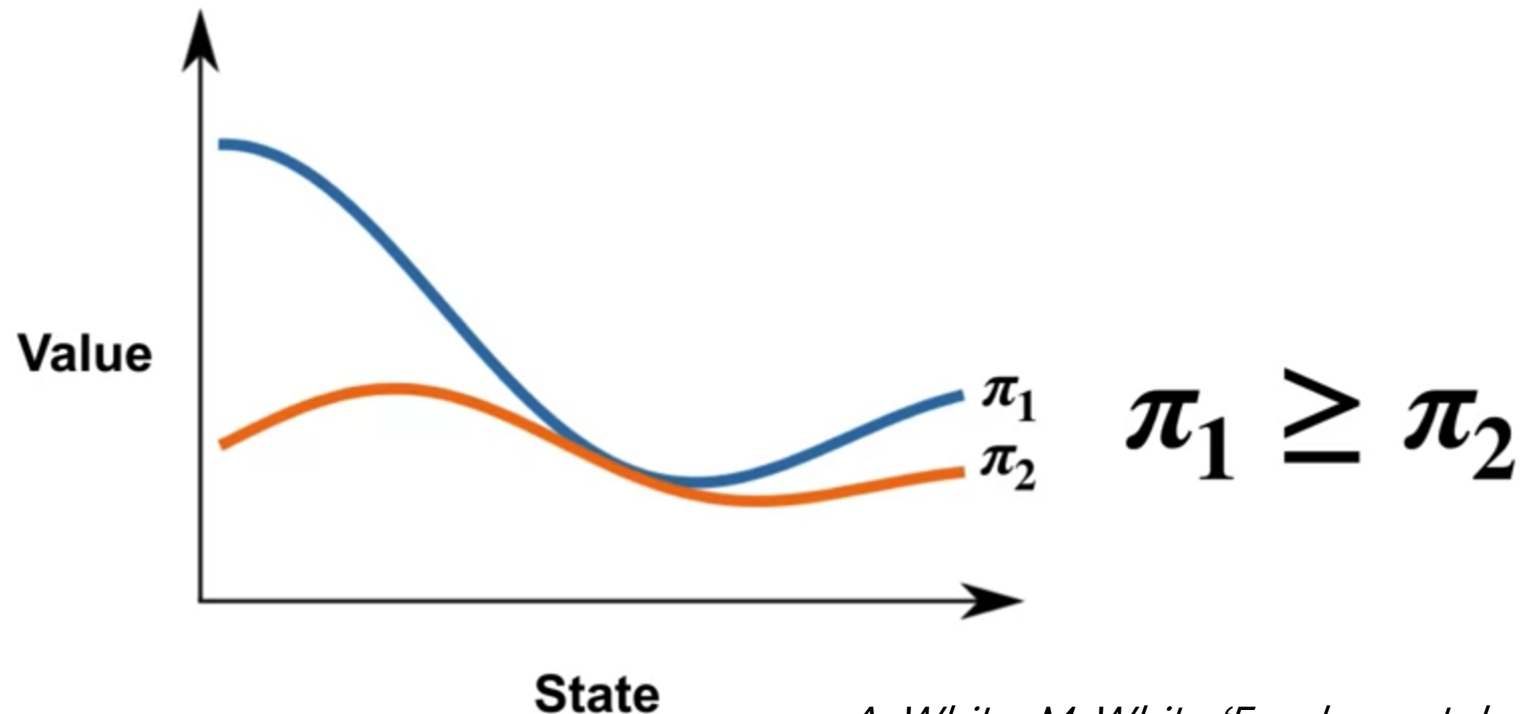
Optimal **action** value function on the Student Markov Chain



Optimal Policy

Our final goal is to find an optimal policy: the best way to act in a MDP!

We define an **order** over policies: $\pi \geq \pi'$ if $v_\pi(s) \geq v_{\pi'}(s)$ for all s



Optimal Policy

Theorem

For any MDP:

1. There exists an optimal policy π_* such that $\pi_* \geq \pi$ for all possible π
2. All optimal policies achieve the optimal value function

$$v_{\pi_*}(s) = v_*(s)$$

3. All optimal policies achieve the optimal value function

$$q_{\pi_*}(s, a) = q_*(s, a)$$

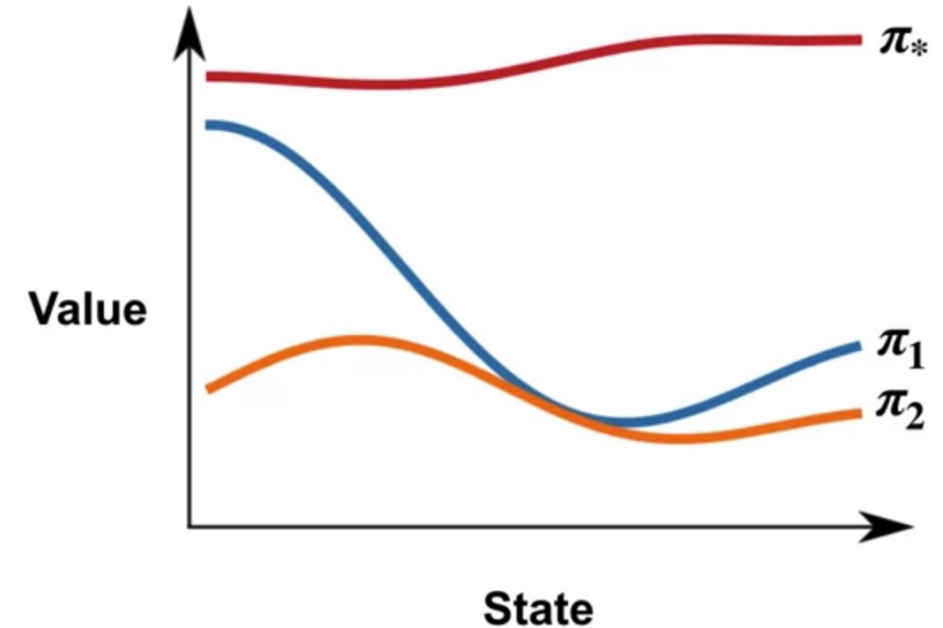
Finding an Optimal Policy

- How to operationally find an optimal policy?

A simple approach, if we have $q_*(s, a)$, if by maximizing over:

$$\pi_*(a|s) = \begin{cases} 1 & \text{if } a = \operatorname{argmax}_{a \in \mathcal{A}} q_*(s, a) \\ 0 & \text{otherwise} \end{cases}$$

- There is always a deterministic optimal policy for any MDP

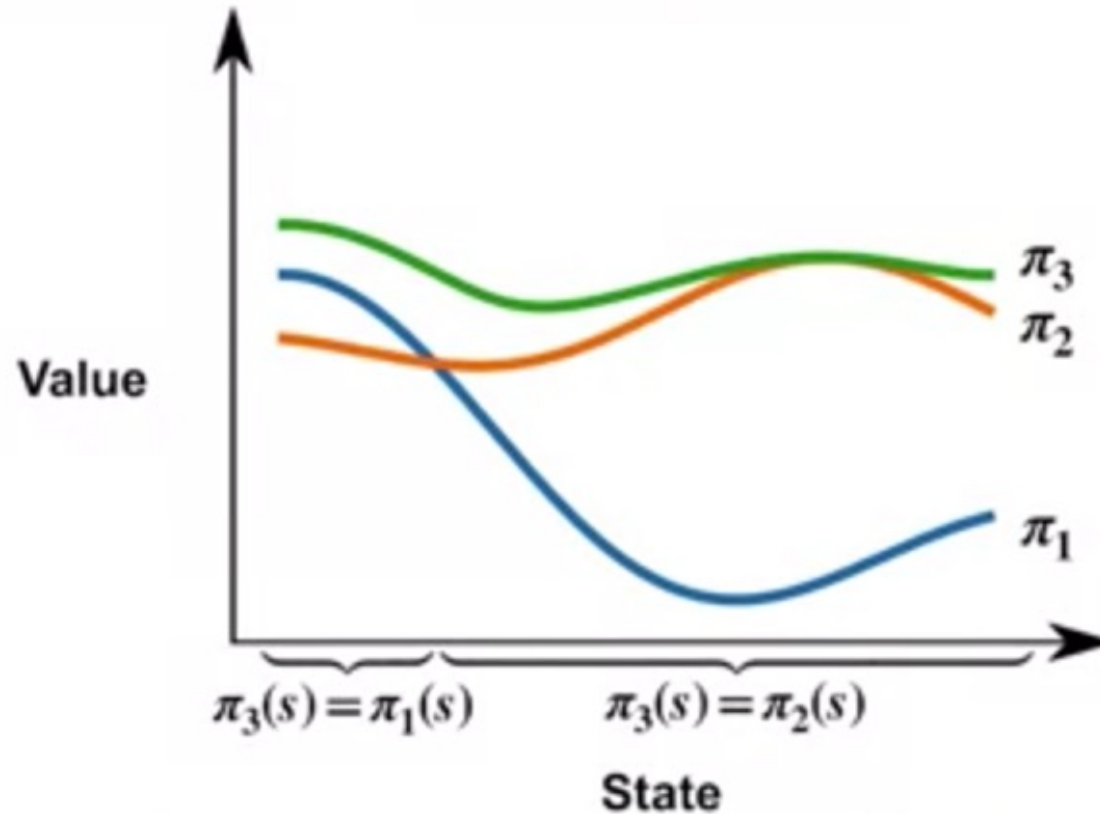


Optimal Policy Existence

- Why there is always an optimal policy?

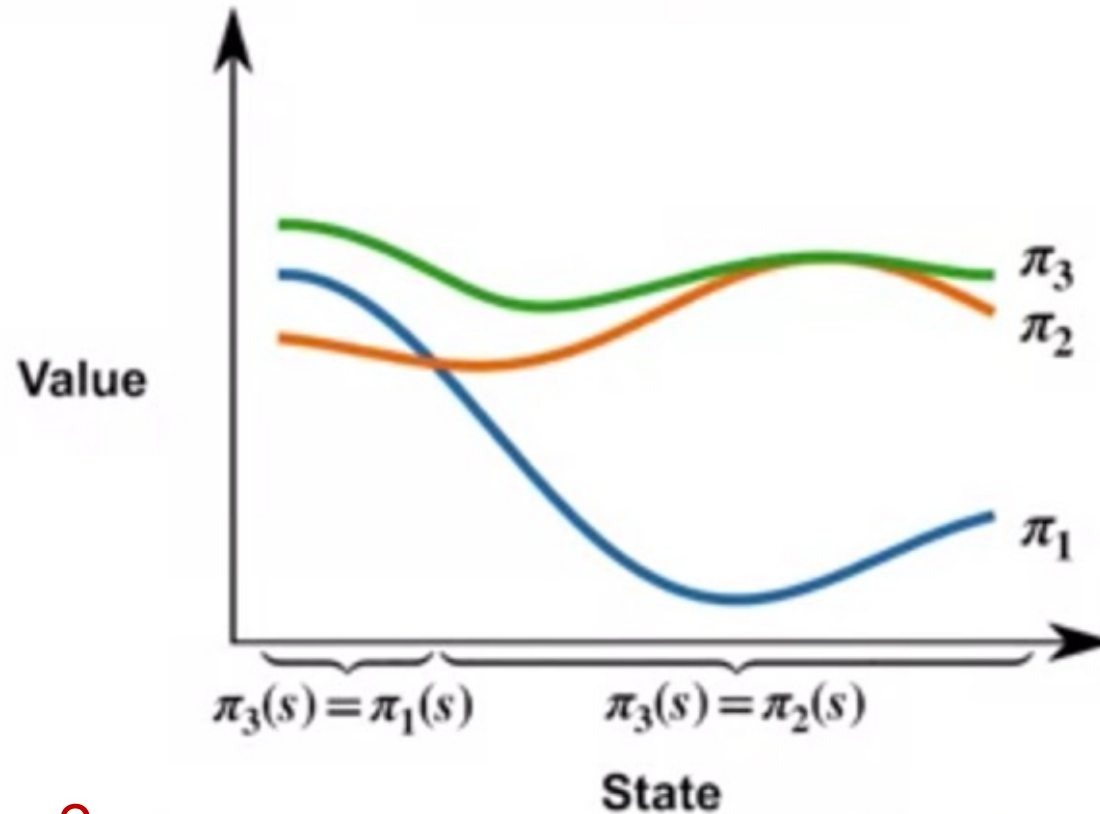
Optimal Policy Existence

- Why there is always an optimal policy?



Optimal Policy Existence

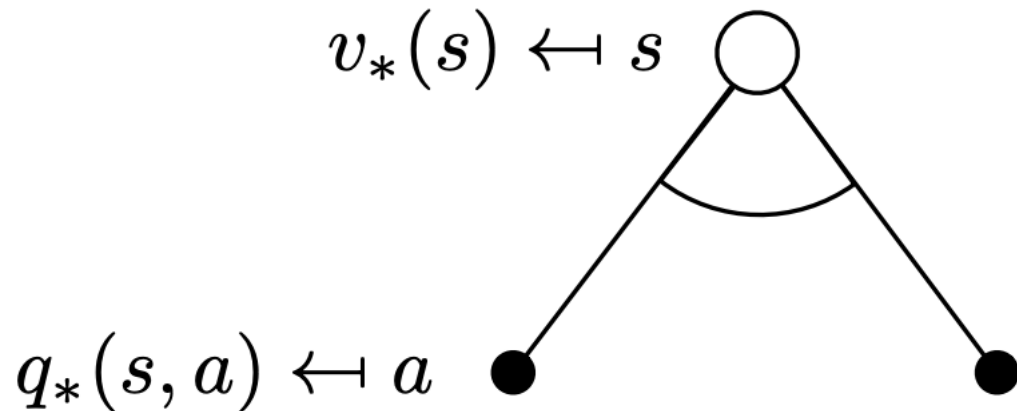
- Why there is always an optimal policy?



But how do we get q_* ?

Bellman Optimality Equation 1/3

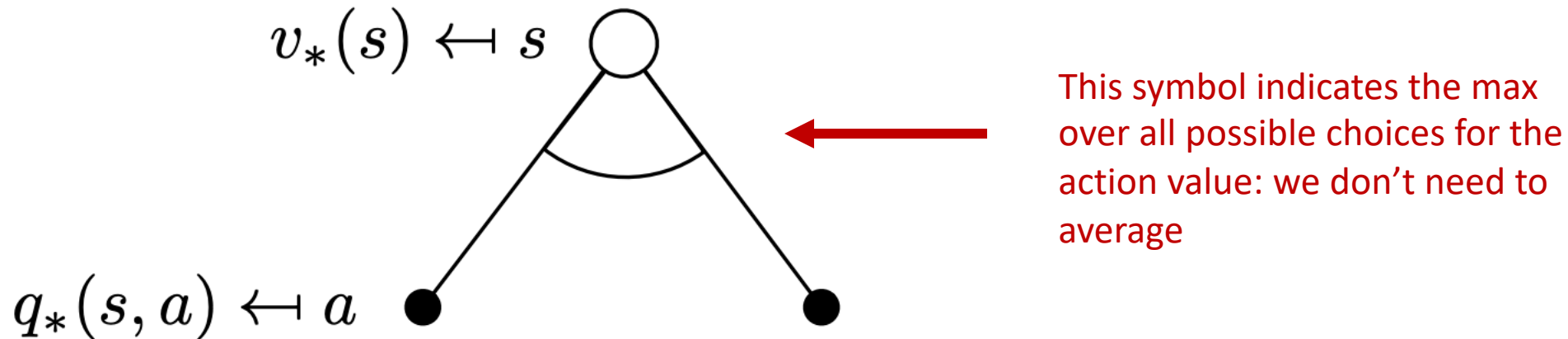
- Not to be confused with the Bellman Expectation Equation, that holds for a generic policy and it is a way to recursively define v_π and q_π
- The Bellman Optimality Equation is a way to define the optimal v_* with itself: we exploit again the 1-step look-ahead principle



$$v_*(s) = \max_a q_*(s, a)$$

Bellman Optimality Equation 1/3

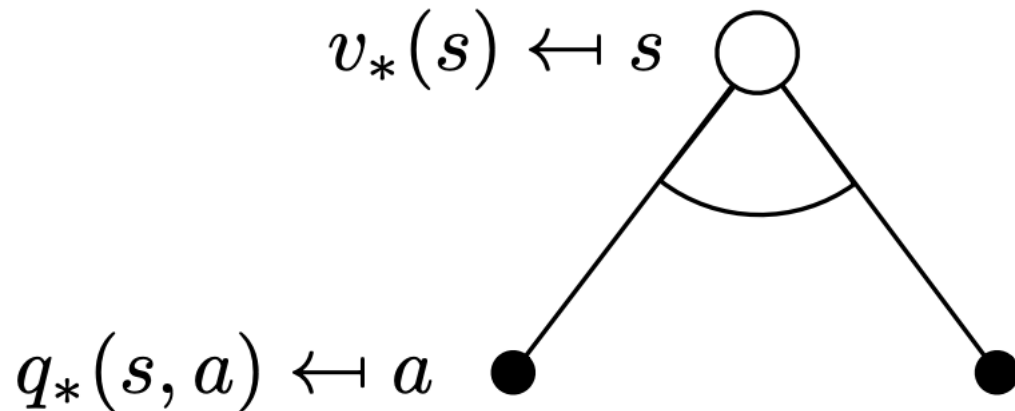
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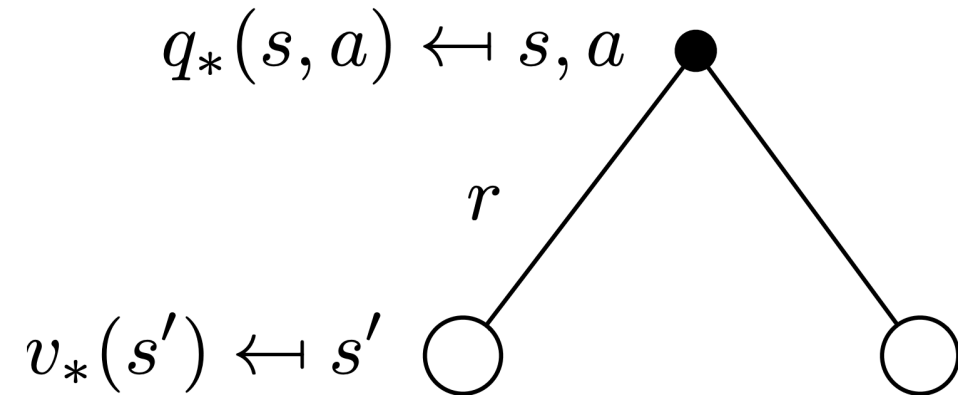
Bellman Optimality Equation 1/3

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- The Bellman Optimality Equation is a way to define the optimal v_* with itself: we exploit again the 1-step look-ahead principle



Agent: here we pick the
'best' action

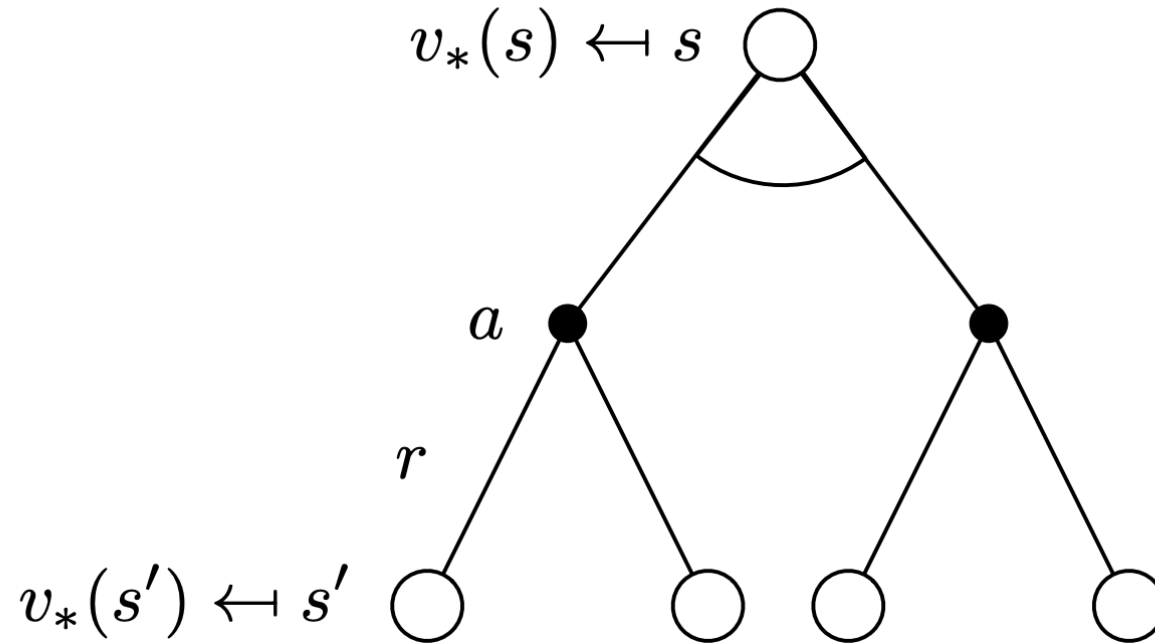
$$v_*(s) = \max_a q_*(s, a)$$



Environment: now
we need to average!

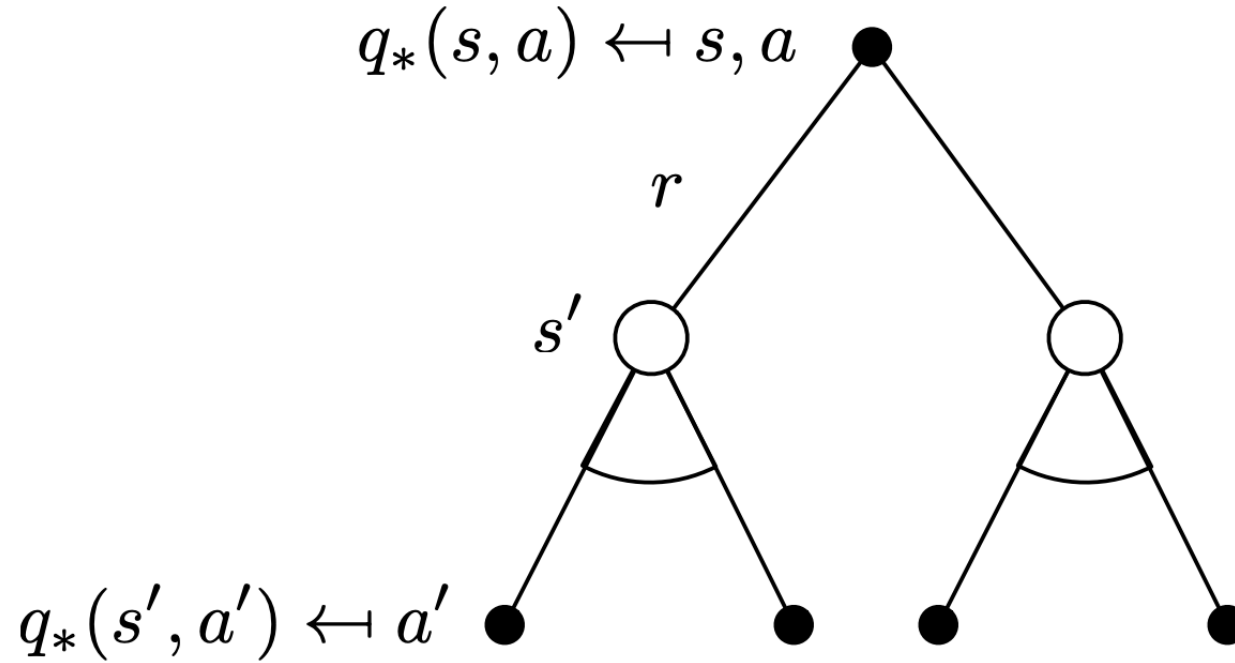
$$q_*(s, a) = \mathcal{R}_s^a + \gamma \sum_{s' \in \mathcal{S}} \mathcal{P}_{ss'}^a v_*(s')$$

Bellman Optimality Equation 2/3



$$v_*(s) = \max_a \mathcal{R}_s^a + \gamma \sum_{s' \in \mathcal{S}} \mathcal{P}_{ss'}^a v_*(s')$$

Bellman Optimality Equation 3/3

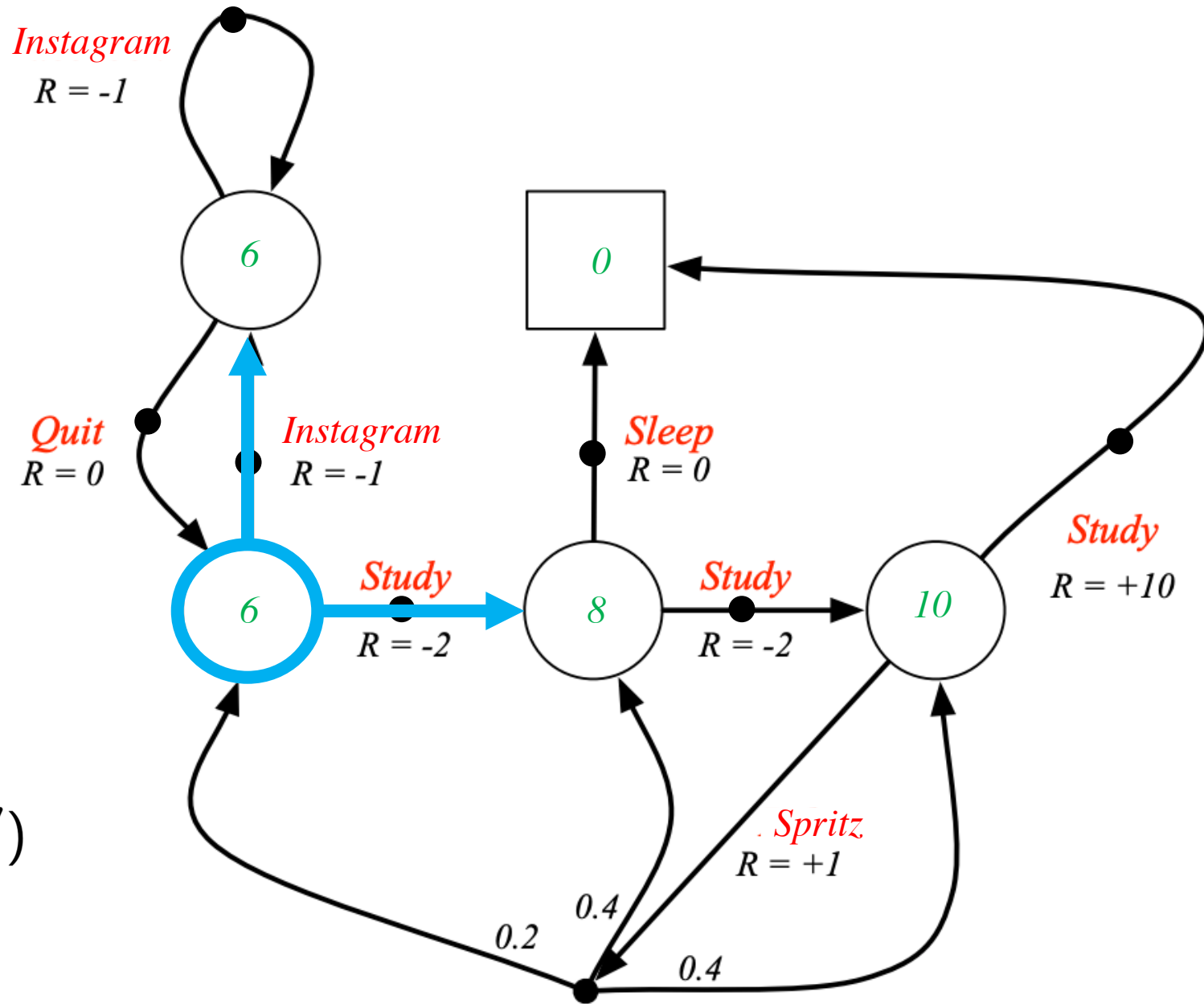


$$q_*(s, a) = \mathcal{R}_s^a + \gamma \sum_{s' \in \mathcal{S}} \mathcal{P}_{ss'}^a \max_{a'} q_*(s', a')$$

Bellman Optimal equation in the Student MDP

$$v_*(s) = \max_a \mathcal{R}_s^a + \gamma \sum_{s' \in \mathcal{S}} \mathcal{P}_{ss'}^a v_*(s')$$

$$6 = \max \{-2 + 8, -1 + 6\}$$



Bellman Optimality Equation

$$v_*(s) = \max_a \mathcal{R}_s^a + \gamma \sum_{s' \in \mathcal{S}} \mathcal{P}_{ss'}^a v_*(s')$$

- Since there is a non-linear operation (a max) we don't have a closed-form solution
- We will resort to **iterative methods**: value iteration, policy iteration, q-learning, SARSA, ...

Summarizing

Markov Decision Processes (MDPs) formally describe an environment for Reinforcement Learning

- i. Markov Processes $\langle \mathcal{S}, \mathcal{P} \rangle$
- ii. Markov Reward Processes $\langle \mathcal{S}, \mathcal{P}, \mathcal{R}, \gamma \rangle$
- iii. Markov Decision Processes (MDPs) $\langle \mathcal{S}, \mathcal{A}, \mathcal{P}, \mathcal{R}, \gamma \rangle$

From now on we will deal with MDPs!

Markov Reward Processes and MDPs can be ‘solved’ with respect to:

- Deriving the value function and the action-value function (in MDP w.r.t. a policy)
- Finding the best policy

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From now on we will deal with MDPs!

Markov Reward Processes and MDPs can be ‘solved’ with respect to:

- Deriving the value function and the action-value function (in MDP w.r.t. a policy) -> (POLICY) EVALUATION
- Finding the best policy -> CONTROL / POLICY IMPROVEMENT



Summarizing

Bellman equations are our tool to 'solve' Markov Reward Processes (MRPs) and MDPs thanks to their recursive nature:

MRP	Bellman equation: for finding value functions	Linear: we can use it for 'small' MRPs. We need to resort to iterative approaches for 'large' MRPs
MDP	Bellman expectation equation: for finding value functions and action-value functions	Linear: we can use it for small MDPs. We need to resort to iterative approaches for 'large' MDPs
MDP	Bellman optimality equation: for finding optimal value functions and optimal action-value functions	Non-linear: we need iterative approaches even for small MDPs.



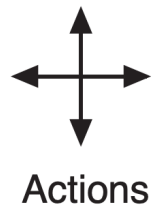
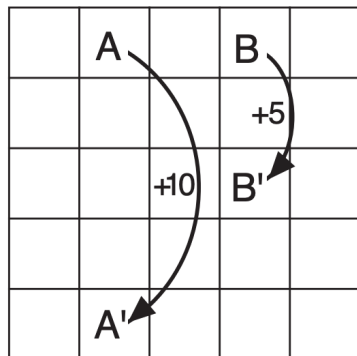
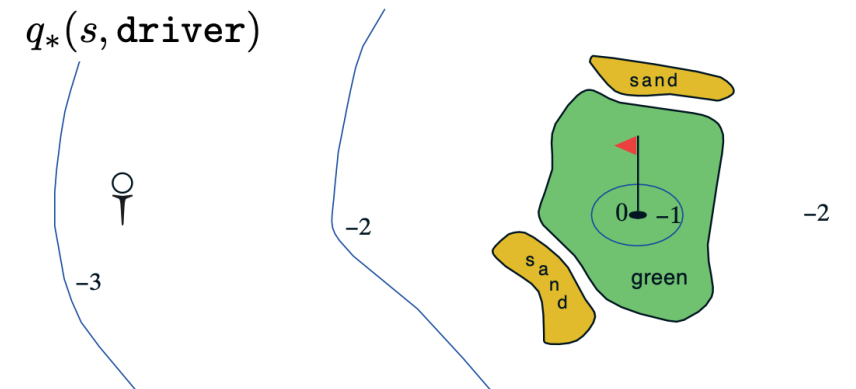
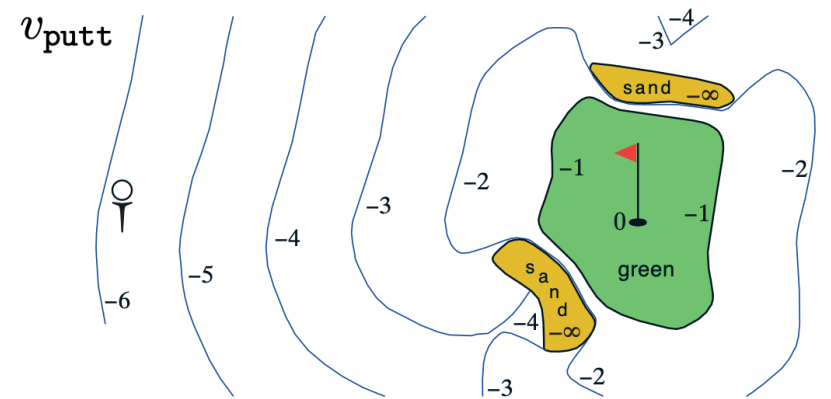
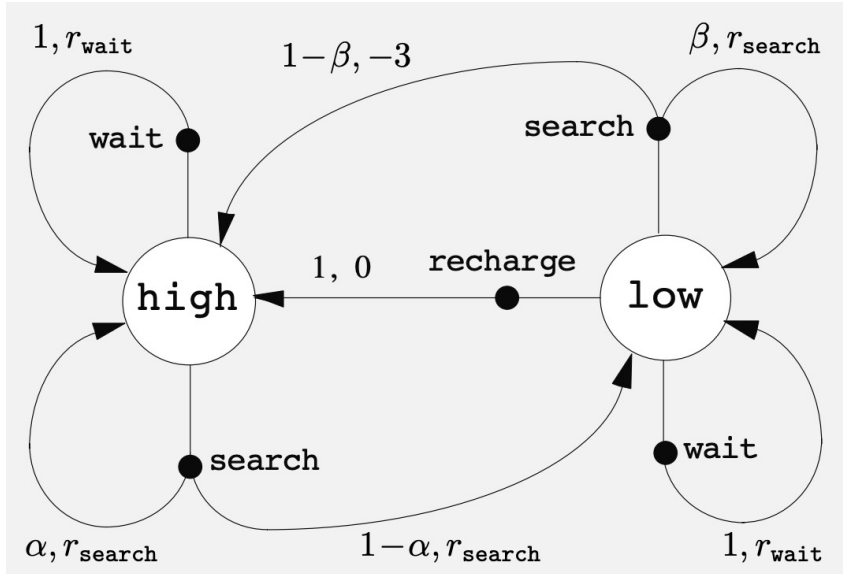
In the book

Summarizing

Bellman equations are our tool to 'solve' Markov Reward Processes (MRPs) and MDPs thanks to their recursive nature:

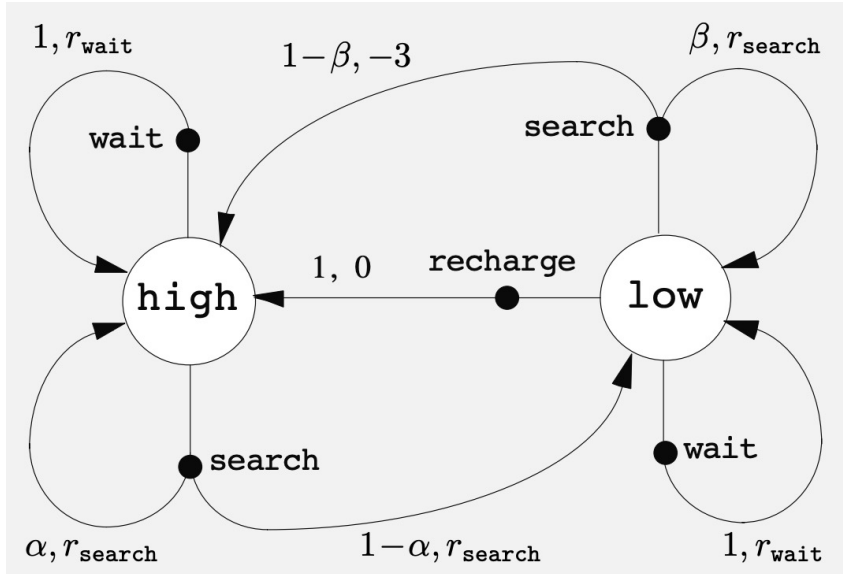
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MDP	Bellman optimality equation: for finding optimal value functions and optimal action-value functions	Non-linear: we need iterative approaches even for small MDPs.

Examples in the Book (chapter 3)

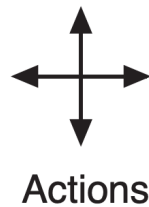
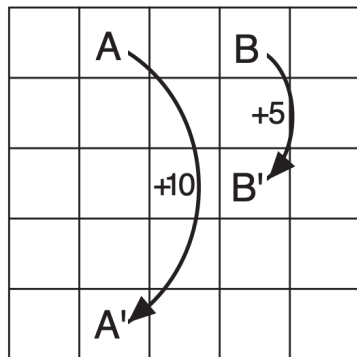
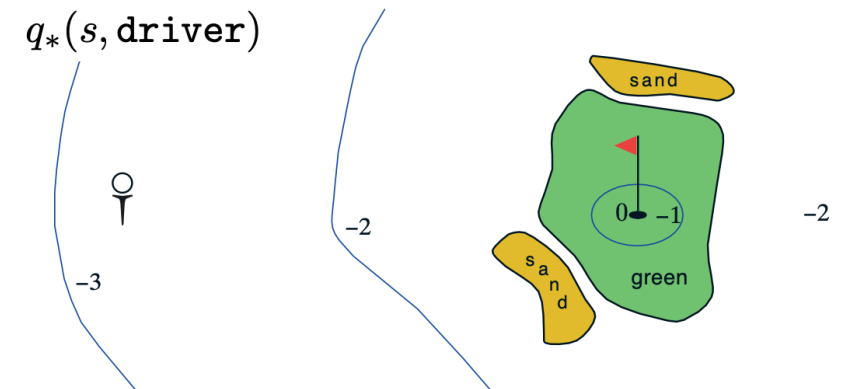
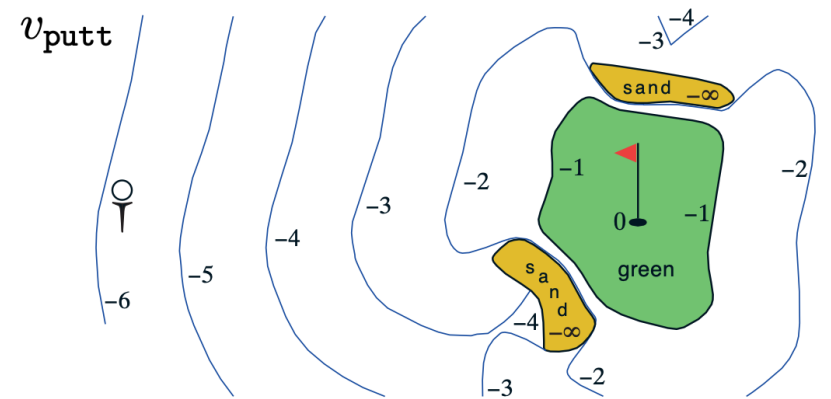


3.3	8.8	4.4	5.3	1.5
1.5	3.0	2.3	1.9	0.5
0.1	0.7	0.7	0.4	-0.4
-1.0	-0.4	-0.4	-0.6	-1.2
-1.9	-1.3	-1.2	-1.4	-2.0

Examples in the Book (chapter 3)



We will not see these in the laboratory (we will move directly to Chapter 4 examples): we suggest to take a look by yourselves!



3.3	8.8	4.4	5.3	1.5
1.5	3.0	2.3	1.9	0.5
0.1	0.7	0.7	0.4	-0.4
-1.0	-0.4	-0.4	-0.6	-1.2
-1.9	-1.3	-1.2	-1.4	-2.0

MDP: Exam

- All the content of Chapter 3 of the book may be exam material
- Keep in mind that the book presents directly MDPs: use these slides for a definition of Markov Processes and Markov Reward Processes

Credits

- Image of the course is taken from C. Mahoney 'Reinforcement Learning' <https://towardsdatascience.com/reinforcement-learning-fda8ff535bb6>
- Overall structure of the lecture and some content was inspired/adapted from D. Silver RL course @ UCL

Thank you! Questions?

Lecture #04 Markov Decision Processes & Bellman Equations

Gian Antonio Susto

